Enterprise Information Systems (Elective)

V Semester SEE Question Papers

Course codes: IM255TBB, IM355TBB, 21IM55B2

- Feb/Mar 2026 (IM255TBB)
- Feb/Mar 2025 (IM355TBB)
- Feb/Mar 2024 (21IM55B2)

Question Paper ID: 750439

RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

Course Code: IM255TBB

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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Enterprise Information Systems

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	List two fundamental ways an Enterprise Information System contributes to improving an organization's operational efficiency.	02	1	1
1.2	Define the term 'Digital Transformation' (DT).	02	1	1
1.3	Define Robotic Process Automation (RPA).	02	2	1
1.4	Name any two leading commercial software platforms (tools) used for implementing RPA.	02	2	1
1.5	Identify a major challenge or pitfall an organization faces when relying on data from multiple, non-integrated systems (data silos) for decision-making.	02	2	2
1.6	State how data from an Enterprise Information System (EIS) supports Strategic level decision-making.	02	2	2
1.7	Define Product Life Cycle Management (PLM) in relation to product development.	02	1	1
1.8	How does the implementation of an integrated EIS/ERP system (like the one used in the Sanyo case) achieve "process quality improvement"?	02	1	2
1.9	State how the Agile practice of "continuous stakeholder feedback" benefits EIS projects regarding user satisfaction.	02	4	2
1.10	Contrast the approach to handling changing requirements between the traditional Waterfall method and the Agile method in an EIS implementation.	02	4	2

Part B

Question No	Question	M	CO	BT
2a	Explain the concept of Digital Transformation (DT). Describe the crucial role of two key technologies, Cloud Computing and Artificial Intelligence (AI), in driving this transformation within a modern enterprise.	08	1	2
2b	Describe the hierarchical role of an Enterprise Information System (EIS) in supporting decision-making across the three organizational management levels: Strategic, Tactical, and Operational.	08	1	2

3a	A manufacturing company's procurement process is complex. Apply three different methods of BPA to the procurement process (from requisition to payment). Justify how each method specifically enhances efficiency in that stage.	08	2	3
3b	A global bank wants to automate its Customer Interaction Process. Design a BPA solution that combines Customer Interaction Automation and Decision Automation. Detail the steps involved and the types of tools necessary for this solution.	08	2	3
	OR			
4a	Explain the role of RPA frameworks such as BPMN and Agile in the lifecycle of an RPA project. Why are these structured approaches necessary even when dealing with tactical automation?	08	2	2
4b	Explain the defining difference between Basic RPA and Advanced RPA (Cognitive RPA). Discuss how Cognitive RPA, through its integration with AI/ML, enables organizations to handle complex tasks that Basic RPA cannot.	08	2	2
5a	A retail chain is analyzing sales data. They observe that sales of product 'A' dropped significantly last month. Apply the four types of data analytics sequentially to investigate this issue and formulate a management response.	08	3	3
5b	A bank wants to deploy an analytical model to predict the likelihood of default for new credit card applicants. Identify the type of data analytics involved and detail the key steps a machine learning model would take to execute this analysis.	08	3	3
	OR			
6a	Explain how the integrated data from an Enterprise Information System (EIS) supports managerial decision-making across the three levels: Operational, Tactical, and Strategic.	08	3	2
6b	Analyze the critical challenge of data silos in an organization. Explain how non-integrated data silos negatively impact the use of data for reliable, cross-functional decision-making.	08	3	2
7a	Describe the organizational conflict or hindrance that can occur when an internally optimized, non-integrated ERP system attempts to coordinate processes with external supply chain partners.	08	1	2
7b	Explain why a fully integrated EIS/ERP system is considered a foundational prerequisite before deploying advanced SCM planning applications. What core issue does the ERP solve that SCM relies upon?	08	1	2
	OR			
8a	Explain how PLM extends beyond traditional product data management by integrating data from upstream (R&D) and downstream (Service/Disposal) stages of the life cycle.	08	1	2
8b	Describe the role of a central EIS/PLM system in facilitating concurrent engineering and collaborative product development. How does this system overcome the sequential delays of traditional design processes?	08	1	2
9a	A business process is undergoing rapid change during a critical ERP customization phase. Analyze the difficulty in handling this change under a strict Waterfall methodology and demonstrate how the Agile principle of "Responding to change over following a plan" provides a superior solution.	08	4	3
9b	Prescribe how the Agile practice of using User Stories and managing a Product Backlog can effectively translate the complex, ambiguous requirements of multiple business departments into clear, actionable development tasks for a new CRM system customization.	08	4	3
	OR			
10a	Explain the importance of Cybersecurity for business-critical EIS (ERP, CRM, SCM). Detail the three foundational objectives of Information Security, known as the CIA Triad. (Sketch Required)	08	4	1
10b	Define and differentiate between four common cybersecurity threats specific to EIS: Phishing, Insider Threat, Data Breach, and Ransomware.	08	4	1

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Regular Examinations Feb/Mar-2025

Industrial Engineering and Management

ENTERPRISE INFORMATION SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables permitted.

PART-A

			M	BT	CO
1	1.1	List two key features of a Customer Relationship Management (CRM) system.	02	2	1
	1.2	Mention two types of data commonly handled by EIS.	02	2	1
	1.3	Define Business Process Automation(BPA).	02	2	1
	1.4	Mention one advantage of implementing automation in business processes.	02	1	1
	1.5	What are the three types of data analytics?	02	2	2
	1.6	Define Product Lifecycle Management (PLM).	02	2	2
	1.7	How does PLM integrate with SCM?	02	2	2
	1.8	How does a web-enabled EIS improve customer satisfaction in SCM?	02	2	2
	1.9	Name two common tools used in Robotic Process Automation (RPA).	02	1	1
	1.10	What is a key characteristic of structured data?	02	2	2

PART-B

2	a	Analyze the benefits and challenges of implementing an EIS in a manufacturing organization. Support your answers with examples.	08	2	1
	b	How can information technology support company's business processes and decision making and give it competitive advantage? Give examples to illustrate your answers.	08	3	1
3	a	A mid-sized manufacturing company faces delays in order processing due to repetitive manual tasks such as data entry and validation. Propose a solution using Robotic Process Automation (RPA), Discuss how implementing RPA can streamline the process and improve efficiency.	08	3	3
	b	A retail company plans to automate its invoice processing system using RPA. Explain the steps involved in the RPA implementation process for this scenarios.	08	2	3
		OR			
4	a	Design a simplified approach to integrate Business Process Automation (BPA) in the inventory management process of an e-commerce company. Include key steps and expected outcomes.	08	3	3

	b	Discuss the importance of incorporating best practices in Business Process Reengineering (BPR) initiatives. Explain how organizations can identify and implement best practices to ensure successful BPR outcomes. Provide examples of best practices commonly used in BPR projects and their impact on business performance.	08	2	3
5	a	A retail chain plans to improve its inventory management system using data warehousing. Explain how a data warehousing can help in managing inventory more effectively. Include key components of a data warehouse and their roles.	08	4	2
	b	Explain how data analytics in Enterprise Information System (EIS) can be used to improve the decision-making process for a manufacturing company aiming to reduce operational costs.	08	2	2
		OR			
6	a	Explain the concept of data mining and its role in extracting actionable insights from large datasets within Enterprise Information System (EIS). Discuss the various techniques and algorithms commonly used in data mining and provide examples of how they can be applied in different business contexts.	08	4	2
	b	Discuss the significance of Customer Relationship Management (CRM) systems in Enterprise Information System (EIS) development. Explain how CRM systems facilitate customer interactions, data analysis and relationship management across various touchpoints. Provide examples of CRM implementation strategies and their impact on enhancing customer satisfaction and loyalty.	08	3	2
7	a	Explain the role and magnitude of Enterprise Information System (EIS) in Supply Chain Management (SCM). How does EIS enhance the efficiency of supply chain operations?	08	2	3
	b	Discuss the impact of web-enabled EIS/ERP systems on CM . How do these systems improve agility and responsiveness in supply chains?	08	3	3
		OR			
8	a	Compare and contrast Enterprise Information System (EIS) and Supply Chain Management (SCM) systems. How are they integrated and what unique roles do they play?	08	3	3
	b	"Product life cycle management augments communication and teams across individual departments within a company". How are they integrated and what unique roles do they play?	08	2	3
9	a	Discuss the advantages and challenges of adopting cloud-based Enterprise Information System (EIS). How does cloud technology enhance enterprise operations?	08	4	4
	b	Provide an overview of Agile project management in Enterprise Information System (EIS). How does Agile methodology improve system implementation?	08	2	4
		OR			
10	a	"Specialized information is too diverse and even redundant". Illustrate and analyze the statement.	08	4	4
	b	Discuss the role of cyber security in safeguarding Enterprise Information System (EIS) against potential threats.	08	2	4

21IM55B2

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Examinations Apr/May-2024

Industrial Engineering and Management

ENTERPRISE INFORMATION SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	_____ Characteristics is typically associated with an Enterprise Information System (*EIS*).	01	1	1
	1.2	_____ is the main benefit of EIS from a technical standpoint.	01	2	2
	1.3	What is the primary purpose of Material Requirements Planning (*MRP*)?	01	1	2
	1.4	List a common challenge faced during *ERP* implementation related to software customization.	02	2	1
	1.5	What is one of the risks associated with Business Process Reengineering?	01	2	2
	1.6	_____ level of process quality design focuses on optimizing individual tasks within a process.	01	1	1
	1.7	Write the full form of *BPM*.	01	2	1
	1.8	How does reengineering differ from incremental improvements?	01	1	2
	1.9	Enumerate the components of Business Intelligence System architecture.	02	2	1
	1.10	Which component of *EIS* provides a visual representation of key performance indicators and metrics related to the supply chain?	01	1	2
	1.11	_____ technology is often used to enhance the visibility and traceability of products in the supply chain.	01	1	1
	1.12	How does the integration of Enterprise Information Systems with the Supply Chain Network contribute to sustainability?	02	2	2
	1.13	Which technology is commonly used to track the movement of goods in real-time within the Supply Chain Network?	01	2	1
	1.14	What is a key benefit of the trend towards a reduction in implementation time in *EIS* projects?	01	1	2
	1.15	How does the adoption of open-source technologies impact Enterprise Information System (*EIS*) implementation?	01	1	2
	1.16	What does *MRP*-3 (Money Resource Planning) primarily focus on?	01	2	1
	1.17	What is a key feature of the progression aspect in next-generation *EIS* software?	01	2	1

PART-B

2	a	Sketch the demand-supply chain service provider model. Explain the concepts of Enterprise Information Systems (*EIS*).	08	2	2
	b	Provide a summary of the features of various Enterprise Resource Planning (*ERP*) modules.	08	2	1

3	a	Develop the typical flow of inputs converted to outputs through a process. Chose a process and list the alternative accomplishing the same. Design the best process.	08	3	4
	b	Enumerate the typical infrastructural processes in an organization. Cite examples.	08	2	2
		OR			
4	a	Identify the common myths about reengineering. Elucidate the conditions under which reengineering works best.	08	3	3
	b	Provide a comparative analysis of clean state and technology enabled reengineering.	08	3	3
5	a	Develop and sketch the roadmap to embrace business intelligence. Summarize the four stages in business intelligence development.	08	4	4
	b	Select three business intelligence software solution vendors and evaluate the performance features.	08	4	4
		OR			
6	a	Write a detailed note on data warehousing. Illustrate the data warehouse linkages with information and business intelligence.	08	2	2
	b	Analyze the importance of online analytical processing ($OLAP$) in business intelligence systems.	08	3	4
7	a	Critically evaluate the statement "EIS applications help in effectively delivering Supply Chain Management".	08	3	3
	b	Explain the role of web enabled Enterprise Information Systems on Supply Chain Management.	08	2	2
		OR			
8	a	Compare Enterprise Information Systems (EIS) and Supply Chain Management (SCM).	08	3	3
	b	"Product life cycle management (PLM) facilitates Enterprise Information Systems and Supply Chain Management systems integration". Illustrate and critically analyze the statement.	08	4	5
9	a	Summarize the paradigm shift across generations of Enterprise Information System software.	08	3	3
	b	Discuss how new technologies enable Enterprise Information System implementation.	08	2	1
		OR			
10	a	Elaborate the following futuristic trends in Enterprise Information System. I) Expenditure trends II) Reduction in implementation time.	08	2	5
	b	Discuss the systems development of tailored processes for newer Enterprise Information System software.	08	2	5

Advanced Decision Modelling (Elective)

V Semester SEE Question Papers

Course codes: IM255TBD, IM355TBD, 18IM52

- Feb/Mar 2026 (IM255TBD)
- Feb/Mar 2025 (IM355TBD)
- Mar/Apr 2023, Decision Sciences II: Stochastic Models (18IM52)
- Jan/Feb 2021, Decision Sciences II: Stochastic Models (18IM52)

Question Paper ID: 923176

Course Code: IM255TBD

RV College of Engineering®
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RV College of Engineering

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R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

INDUSTRIAL ENGINEERING AND MANAGEMENT

Advanced Decision Modelling

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	State the behaviors of customer in a queuing model?	02	1	2
1.2	When arrival is 3 per hour and service is 5 per hour in an M/M/1 queue, the probability of 3 persons being in the system is _____	02	1	2
1.3	What is an absorbing state?	02	1	2
1.4	Classify the states of Markov Chain.	02	1	2
1.5	What is the objective of inventory problem analysis?	02	2	2
1.6	What is the main meaning/purpose of inventory control?	02	2	2
1.7	What is the main purpose of replacement analysis?	02	2	2
1.8	State the applications of Applications of Replacement and Maintenance Models	02	2	2
1.9	What are the standard assumptions in classical sequencing problems?	02	4	2
1.10	What is Johnson's Rule?	02	4	2

Part B

Question No	Question	M	CO	BT
2a	Consider a box office ticket window being manned by a single server. Customer arrives to purchase ticket according to Poisson input process with a mean rate of 30/hr. the time required to serve a customer has an ED with a mean of 90 seconds determine: (i) Mean queue length. (ii) Mean waiting time in the system.	08	1	3

	(iii) The probability of the customer waiting in the queue for more than 10min. (iv) The fraction of the time for which the server is busy.			
2b	An M/M/1 queuing system has an average arrival rate of $\boldsymbol{\lambda}$ = 8 customers per hour and an average service rate of μ = 10 customers per hour. i) Check whether the system is stable. ii) Find the average number of customers in the system. iii)Find the average number of customers in the queue. iv)Find the average waiting time in the system.	08	1	3
3a	Explain the classification of states in Markov chains and discuss their practical significance.	08	1	2
3b	Discuss the topological properties of the graphs of the following Markov chains: (a) $P = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ (b) $P = \begin{bmatrix} 0.5 & 0.5 \\ 1 & 0 \end{bmatrix}$ (c) $P = \begin{bmatrix} 1/3 & 0 & 2/3 \\ 0 & 1 & 0 \\ 0 & 1/5 & 4/5 \end{bmatrix}$ (d) $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (e) $P = \begin{bmatrix} 1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$	08	2	2
	OR			
4a	A system has three possible states, 0, 1 and 2. Every hour it makes a transition to a different state, which is determined by a coin flip. For example, from state 0, it makes a transition to state 1 or state 2 with probabilities 0.5 and 0.5. i) Find the transition probability matrix. ii) Find the three-step transition probability matrix. iii) Find the steady-state distribution of the Markov chain.	08	2	3
4b	A retail store analyses the loyalty behaviour of its loyalty-card customers to understand long-term retention patterns. Based on historical data, customers are classified into the following states: • State 1: Regular buyer • State 2: Occasional buyer • State 3: Lost customer Customer movement from one period to the next follows a Markov process.	08	2	3

Initially, without any discount scheme, the one-period transition probability matrix was:

$$P_0 = \begin{bmatrix} 0.6 & 0.3 & 0.1 \\ 0.2 & 0.5 & 0.3 \\ 0 & 0 & 1 \end{bmatrix}$$

To improve customer retention, the store introduces a new discount policy. As a result, the transition probabilities from the Regular Buyer state (State 1) change as follows:

- Probability of remaining a Regular buyer = 0.75
- Probability of becoming an Occasional buyer = 0.20
- Probability of becoming a Lost customer = 0.05

All other transition probabilities remain unchanged.

Questions:

i) Construct the new transition probability matrix after the introduction of the discount scheme.
ii) Using the new transition matrix, determine the expected time before a Regular buyer becomes a Lost customer, assuming the customer starts in State 1.

5a	A firm allows backorders. D=20,000, S=₹500, H=₹10. Optimal Q* with shortage is 1600. Find shortage cost P?	08	3	3
5b	A manufacturing company uses steel bolts regularly. Its annual requirement is 24,000 bolts. Cost of placing one order is ₹500 and holding cost is ₹12 per bolt per year. Find: (i) EOQ (ii) number of orders per year (iii) time between orders (iv) minimum annual ordering + holding cost	08	3	3
	OR			
6a	Given: batteries/year, purchase price = Rs.28, holding cost rate = 30% ⇒ per unit/year, ordering cost . Current order quantity batteries (per month). Calculate Costs for current order quantity, EoQ, Number of orders per year using EoQ and costs for EoQ	08	3	3
6b	A company needs 26,000 units/year, S=₹150, H=₹5. Find EOQ and time between orders (weeks)	08	3	3
7a	A delivery truck costs ₹90,000 with a scrap value of ₹10,000. The running costs for different years are: Year | 1 | 2 | 3 | 4 | 5 Running Cost (₹) | 6,000 | 9,000 | 13,000 | 18,000 | 24,000 Find the year in which the truck should be replaced to minimize average cost.	08	4	3
7b	A factory purchases a generator for ₹2,00,000. The scrap value at the end of its life is ₹40,000. The yearly operating costs are: Year | 1 | 2 | 3 | 4 Running Cost (₹) | 15,000 | 22,000 | 30,000 | 40,000 Determine the economic life of the generator.	08	4	3
	OR			

<table>
<tr><td>8a</td><td>Two types of bulbs are available for a factory: Type A bulb costs ₹20 and has an average life of 1,000 hours. Type B bulb costs ₹30 and has an average life of 2,000 hours. Electricity and other costs are same. Which bulb is more economical?</td><td>08</td><td>4</td><td>3</td></tr>
<tr><td>8b</td><td>A printing press machine is purchased at a cost of ₹1,60,000 and its scrap value is ₹30,000. The annual running expenses are as follows:

<table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>Running Cost (₹)</td><td>14,000</td><td>19,000</td><td>25,000</td><td>32,000</td><td>40,000</td><td>49,000</td></tr></table>
Determine the economic replacement age of the machine.</td><td>08</td><td>4</td><td>3</td></tr>
<tr><td>9a</td><td>The following data pertains to a manufacturing activity.

<table><tr><th>Job, j</th><th>Processing time, t_j</th><th>Weight, w_j</th></tr><tr><td>1</td><td>7</td><td>1</td></tr><tr><td>2</td><td>18</td><td>2</td></tr><tr><td>3</td><td>6</td><td>1</td></tr><tr><td>4</td><td>8</td><td>2</td></tr><tr><td>5</td><td>12</td><td>3</td></tr></table>
Determine the sequence which will minimize the weighted mean flowtime of the above problem. Also find the weighted mean flowtime</td><td>08</td><td>4</td><td>3</td></tr>
<tr><td>9b</td><td>Consider the following two machines and six jobs flow shop-scheduling problem. Using Johnson’s algorithm, obtain the optimal sequence, which will minimize the makespan.

<table><tr><td>Job</td><td>Machine 1</td><td>Machine 2</td></tr><tr><td>1</td><td>5</td><td>4</td></tr><tr><td>2</td><td>2</td><td>3</td></tr><tr><td>3</td><td>13</td><td>14</td></tr><tr><td>4</td><td>10</td><td>1</td></tr><tr><td>5</td><td>8</td><td>9</td></tr><tr><td>6</td><td>12</td><td>11</td></tr></table></td><td>08</td><td>4</td><td>3</td></tr>
<tr><td colspan="5">OR</td></tr>
<tr><td>10a</td><td>Critically examine the assumptions made in sequencing problems with respect to real industrial environments. Suggest practical modifications that can make sequencing models more realistic.</td><td>08</td><td>2</td><td>3</td></tr>
<tr><td>10b</td><td>Analyze the applicability and limitations of Johnson’s rule in real-world scheduling problems. What happens when the assumptions of Johnson’s rule are violated?</td><td>08</td><td>2</td><td>3</td></tr>
</table>

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RV COLLEGE OF ENGINEERING®

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V Semester B. E. Regular Examinations Feb/Mar-2025

Industrial Engineering and Management

ADVANCED DECISION MODELING (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

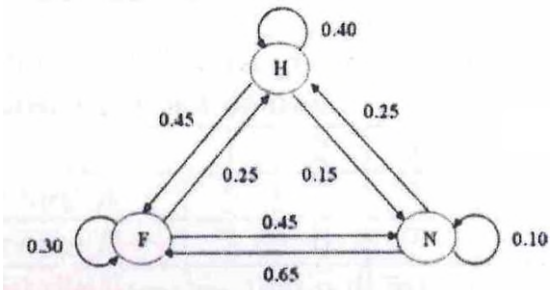
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical tables permitted.

PART-A

			M	BT	CO
1	1.1	Why is queuing theory important in service industries?	02	2	2
	1.2	How does increasing the arrival rate affect the average waiting time in the system?	02	2	2
	1.3	How to the transition rates in a continuous-time Markov chain differ from transition probabilities in a discrete-time Markov chain?	02	2	3
	1.4	What are steady-state probabilities in a Markov chain?	02	2	3
	1.5	What should be considered when applying the EOQ model with quantity discounts?	02	2	2
	1.6	How does the review period affect the deterministic periodic review model?	02	2	3
	1.7	List the key phases of the Bathtub Curve in Replacement and Maintenance Models.	02	2	3
	1.8	What are the two main types of replacement policies?	02	2	2
	1.9	What is the primary objective of sequencing problems? Mention any two.	02	2	2
	1.10	How does the concept of 'setup time' impact sequencing problems?	02	2	2

PART-B

2	a	Explain with a neat diagram, the structure of a queuing system.	06	2	2
	b	In a departmental store one cashier is there to serve the customers. And the customers pick up their needs by themselves. The arrival rate is 9 customers for every 5 minutes and the acshier can serve 10 customers in 5 minutes. Assuming Poisson arrival rate and exponential distribution for service rate, find: i) Average number of customers in the system ii) Average number of customers in the queue or average queue length iii) Average time a customer spends in the system iv) Average time a customer waits before being served.	10	4	4
3	a	What is a discrete-time Markov chain ($DTMC$), and how is it characterized?	08	2	2
	b	State the steps of constructing Matrix of Transition Probabilities of Markov chain.	08	3	3
		OR			

4	a	What is a stochastic process, and how does it differ from a deterministic process?	06	2	2
	b	A professor of statistics not wanting to be predictable, decided on an innovative way of assigning homework based on probabilities. On the first day of the week, he draws a transition diagram as shown in the following Fig. 4.b. The nodes of the diagram represent full credit (F), half credit (H) and no credit (N) assignments. The transition probabilities for the day are as shown in figure. The initial probability distribution of the three state is $(0.3, 0.4, 0.3)$.  Fig. 4.b i) Construct the transition probability matrix ii) $P(X_2 = H)$	10	3	3
5	a	A particular item has a demand of 9000 units per year. The cost of one procurement is $Rs.100$ and the holding cost unit is $Rs.2.40$ per year. The replacement is instantaneous and no shortages are allowed. Determine: i) The economic lot size ii) The number of orders per year iii) The time between orders iv) The total cost per year if the cost of one unit is $Re.1$.	10	3	2
	b	List the limitation of using Economic Order Quantity.	06	2	3
		OR			
6	a	An item is produced at the rate of 50 items per day. The demand occurs at the rate of 25 items per day. If the setup cost is $Rs.100$ per setup and holding cost is $Rs.0.01$ per unit of item per day, find the economic lot size for one run, assuming the shortages are not allowed. Also find the time of cycle and minimum total cost for one run.	08	3	3
	b	A newspaper boy buys paper for 0.05 paise each and sells them for 0.06 paise each. He cannot return unsold newspapers. Daily demand 'r' for newspapers follows the distribution: $Demand\ 'r'$: 10 | 11 | 12 | 13 | 14 | 15 | 16 $Probability\ p(r)$: 0.05 | 0.15 | 0.40 | 0.20 | 0.10 | 0.05 | 0.05 If each day's demand is independent of the previous day's demand, how many papers should be ordered each day?	08	3	3
7	a	Explain the following sudden failure of replacement models with graphs i) Progressive Failure ii) Retrogressive Failure iii) Random Failure	06	2	2

	b	The data collected in running a machine, the cost of which is $Rs.60{,}000$ are given below: $Year$: 1 | 2 | 3 | 4 | 5 $Resale\ value\ (Rs)$: 42,300 | 30,000 | 20,400 | 14,400 | 9,650 $Cost\ of\ spares\ (Rs)$: 4,000 | 4,270 | 4,880 | 5,700 | 6,800 $Cost\ of\ labour\ (Rs)$: 14,000 | 16,000 | 18,000 | 21,000 | 25,000 Determine the optimum period for replacement of Machine.	10	3	3
		OR			
8	a	Discuss the importance of replacement models.	02	2	2
	b	The percentage failure of bulbs at the end of weeks is given below. The cost of replacing individual bulbs $Re.1$. The decision is taken to replace all bulbs simultaneously at fixed intervals and also to replace individual bulbs as they fail in service. If the cost of group replacement is 0.35, find which policy is to be adopted, if initially there are 100 bulbs. $Week$: 1 | 2 | 3 | 4 | 5 $Percentage\ failing\ by\ the\ end\ of\ week$: 10 | 25 | 50 | 80 | 100	14	3	3
9	a	There are five jobs, each of which must go through the two machines A and B in the order AB. Processing times are given below: Job | 1 | 2 | 3 | 4 | 5 $Machine\ A$ | 5 | 1 | 9 | 3 | 10 $Machine\ B$ | 2 | 6 | 7 | 8 | 4 Determine a sequence for the five jobs that will minimize the total elapsed time.	08	3	4
	b	We have five jobs, each of which must go through the machine A, B and C in the order ABC. Determine the sequence that will minimize the total elapsed time. $Job\ No$ | 1 | 2 | 3 | 4 | 5 $Machine\ A$ | 5 | 7 | 6 | 9 | 5 $Machine\ B$ | 2 | 1 | 4 | 5 | 3 $Machine\ C$ | 3 | 7 | 5 | 6 | 7	08	3	4
		OR			
10	a	In the context of manufacturing and production systems, explain role of priority rules related to sequencing problems.	06	2	4
	b	Use the graphical method to minimize time needed to process the following jobs on the machines shown, i.e. for each machine find the job that should be done first. Also, calculate the total elapsed time to complete both jobs. $Machine$ $Job\ 1\ sequence$: A B C D E $Time\ (hrs)$: 3 4 2 6 2 $Machine$ $Job\ 2\ sequence$: B C A D E $Time\ (hrs)$: 5 4 3 2 6	10	3	4

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations March / April-2023

Industrial Engineering and Management

DECISION SCIENCES II (STOCHASTIC MODELS)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

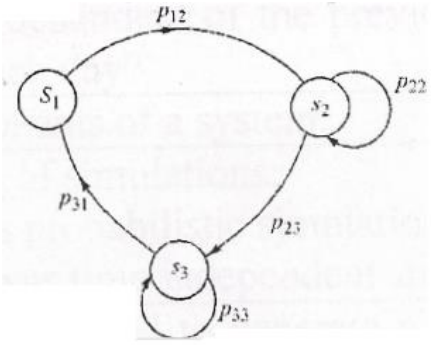
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of Statistical Table is permitted.

PART-A

1	1.1	List the behavior of arrivals in a queuing system.	02
	1.2	What is static and dynamic queue discipline?	02
	1.3	State the purpose of inventory management.	02
	1.4	List the four components of total inventory cost.	02
	1.5	Probabilities of occurrence of any state are representing one of the _____ numbers of states of nature in the _____.	02
	1.6	What will be the number of equilibrium equations would be, if a matrix of transition probability is of the order $n \times n$?	02
	1.7	The Monte-Carlo method of simulation is developed through the use of _____ and _____.	02
	1.8	Validation is the process of _____ a model to the real system that it represents to make sure that it is _____.	02
	1.9	List the applications, other than systems simulation, for pseudo random generations.	02
	1.10	Is Erlang distribution same as gamma distribution?	02

PART-B

2	a	Explain transient state and steady state condition of a queuing system.	06
	b	A warehouse has only one loading dock manned by a three person crew. Trucks arrive at the loading dock at an average rate of 4 trucks per hour and the arrival rate is Poisson distributed, The loading of a truck takes 10 minutes on an average and can be assumed to be exponentially distributed. The operating cost of a truck is Rs.20 per hour and the members of the loading crew are paid Rs.6 each per hour. Would you advise the truck owner to add another crew of three persons?	10

3	a	State the characteristics of Markov Chain	06
	b	Explain with the diagram shown in Fig 3b, transition probability and transition matrix.  Fig 3b	10
		OR	
4	a	Determine whether or not the following matrices could be a transition matrix for a Markov chain. For those that are not, explain why not, and for those that are, draw a picture of the chain. i) $\begin{vmatrix} 0.5 & 0 \\ 0 & 0.5 \end{vmatrix}$ ii) $\begin{vmatrix} 2/3 & 1 \\ 1/3 & 0 \end{vmatrix}$ iii) $\begin{vmatrix} 0.25 & 0.5 \\ 0.75 & 0.5 \end{vmatrix}$	06
	b	A company is considering using Markov theory to analyze brand switching between three different brands of floppy disks. Survey data has been gathered and has been used to estimate the following transition matrix for the probability of moving between brands each month: $$\begin{array}{cc\|ccc} & & \multicolumn{3}{c}{To\ Brand} \\ & & 1 & 2 & 3 \\ \hline & 1 & 0.80 & 0.10 & 0.10 \\ From\ brand & 2 & 0.03 & 0.95 & 0.02 \\ & 3 & 0.20 & 0.05 & 0.75 \end{array}$$ The current (month1) market shares are 45%, 25% and 30% for brands 1, 2 and 3 respectively. i) What will be the expected market shares after two months have elapsed (i.e. in month 3)? ii) What is the long run prediction for the expected market share for each of the three brands?	10
5	a	Give the type of inventory held for the following type of organization: i) Manufacture ii) Hospital iii) Bank iv) Airline company	06
	b	The production department of a company requires 3,600 kg of raw material for manufacturing a particular item per year. It has been estimated that the cost of placing an order is $Rs.36$ and the cost of carrying inventory is 25 percent of the investment in the inventories. The price is Rs.10 per kg.	

Calculate:

i) The optimal lot size
ii) The optimal order cycle time
iii) Per year minimum total inventory variable cost
iv) Per year minimum total inventory cost — 10

OR

6 a The pattern of demand for a seasonal product is as follows:

Demand (in units)	1	2	3	4	5	6	7	8
Probability	0.05	0.10	0.15	0.20	0.20	0.15	0.10	0.05

The cost of product is Rs.80 per unit and the selling price is Rs.120 How many units should be purchased for the season so as to maximize the expected profit? — 06

b A newspaper boy buys papers for 0.05 paise each and sells them for 0.06 paise each. He cannot return unsold newspapers. Daily demand 'r' for newspapers follows the distribution:

Demand 'r'	10	11	12	13	14	15	16
Probability $p(r)$	0.05	0.15	0.40	0.20	0.10	0.05	0.05

If each day's demand is independent of the previous day's demand, how many papers should be ordered each day? — 10

7 a Explain the different components of a system. — 10

b Discuss the following types of simulations:

i) Deterministic versus probabilistic simulation
ii) Time dependent versus time independent simulation — 06

8 a Use the linear congruential method to generate a sequence of three two-digit random integers and corresponding random numbers. Let $X_o = 27, a = 8, c = 47$ and $m = 100$. — 06

b The sequence of umbers $0.54, 0.73, 0.98, 0.11$ and 0.68 has been generated. Use the Kolmogorov- Smirnov test with $\alpha = 0.05$ to learn whether the hypothesis that the numbers are uniformly distributed on the interval $[0, 1]$ can be rejected. — 10

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Examinations Jan/Feb – 2021

Industrial Engineering and Management

DECISION SCIENCE II (STOCHASTIC MODELS)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6
3. Use of Statistical Tables is permitted. (Ks test and ChiSquare tables)

PART-A

1	1.1	In queue designation A/B/S: (d/f), what does S represent?	01
	1.2	The distribution of arrivals in a queuing system can be considered as a _____ process.	01
	1.3	If the operating characteristics of a queue are dependent on time, then it is said to be in a _____ state.	01
	1.4	The characteristics of a queuing model are independent of _____ the queue discipline.	01
	1.5	Write the probability density function of a Markov process.	01
	1.6	In Markov analysis, the likelihood that any system will change from one period to the next is revealed by the matrix of _____ probabilities.	01
	1.7	A problem is classified as Markov chain provided; there are _____ number of possible states.	01
	1.8	The probability that a system is in a particular state after a large number of periods is _____ of the beginning state of the system.	01
	1.9	At Economic Order Quantity, annual ordering cost is _____ to annual carrying cost.	01
	1.10	Carrying cost _____ with quantity ordered per order.	01
	1.11	The ordering cost per order and average unit carrying cost are constant, and demand suddenly falls by 75 % then EOQ will _____ by 50 percent.	01
	1.12	The water consumption from a water tank follows _____ system.	01
	1.13	If EOQ = 40 units, order costs are Rs. 2 per order, and carrying costs are Rs. 20 per unit, what is the usage in units?	01
	1.14	The cost of insurance and taxes are included in _____ carrying cost.	01
	1.15	The time period between placing an order and its receipt in stock is known as _____ time.	01
	1.16	The purpose of using simulation technique is to _____ a real world situation.	01
	1.17	Biased random sampling is made from among alternatives which have _____ probability.	01

	1.18	A simulation uses the logical relationships and mathematical expressions of the _____ system.	01
	1.19	_____ model can only show the values that system attributes take when the system is in balance.	01
	1.20	From a practical perspective, if we have a waiting line problem for which the Poisson and negative exponential distributions do not apply, and we desire a reasonably accurate solution, we should use _____.	01

PART-B

2	a	Discuss the characteristics of a queuing system.	08
	b	A super market has a single cashier. During the peak hours, customers arrive at a rate of 20 customers per hour. The average no of customers that can be processed by the cashier is 24 per hour. Find (i) The average no of customers in the queue system (ii) The average time a customer spends in the system. (iii) The average time a customer spends in queue. (iv) The any time a customer spends in the queue waiting for service	08
3	a	State and discuss the characteristics of a Markov Chain.	06
	b	State the steps of constructing Matrix of Transition Probabilities of Markov Chain.	10
		OR	
4	a	State the application of Markov analysis.	04
	b	A food products' company is contemplating the introduction of a revolutionary new product with packaging or replacing the existing product at much higher price (S1). It may even make a moderate change in the composition of the existing product, with a new packaging at a small increase in price (S2), or may make a small change in the composition of the existing product, backing it with the word 'New' and a negligible increase in price (S3). The three possible states of nature or events are (i) high increase in sales (N1), (ii) no change in sales (N2) and (iii) decrease in sales (N3). The marketing department of the company worked out the payoffs in terms of yearly net profits for each of the strategies of three events (expected sales). This is represented in the following table:	12

		States of Nature		
		N1	N2	N3
Strategies	S1	7,00,000	3,00,000	1,50,000
	S2	5,00,000	4,50,000	0
	S3	3,00,000	3,00,000	3,00,000

Which strategy should the concerned executive choose on the basis of
i. Maximin criterion
ii. Maximax criterion
iii. Minimax regret criterion
iv. Laplace criterion

5 a List the assumptions of economic order quantity model. 06

b ABC Ltd. uses EOQ logic to determine the order quantity for its various components and is planning its orders. The Annual consumption is 80,000 units, Cost to place one order is Rs. 1,200, Cost per unit is Rs. 50 and carrying cost is 6% of Unit cost. Find

i. Economic Order Quantity
ii. No. of orders per year
iii. Ordering Cost
iv. Carrying Cost
v. Total Cost of Inventory. 10

OR

6 a Explain the following cost associated with inventories:

i. Purchase cost
ii. Carrying cost
iii. Ordering cost 06

b A contractor of second hand motor trucks use to maintain a stock of trucks every month. The demand of the trucks occur at a constant rate but not in constant size. The probability distribution of the demand is as shown below:

Demand	0	1	2	3	4	5	6 or more
Probability	0.40	0.24	0.20	0.10	0.05	0.01	0.00

The holding cost of an old truck in stock for one month is /Rs.100/- and penalty for a truck if not delivered to the demand, is Rs. 1000/-. Determine the optimal size of the stock for the contractor. 10

7 a Explain at least four different types of model. 08

b Using Multiplicative Congruential Method, generate sequence of 5 integer numbers where a = 24, m = 100 and seed value = 64. 08

8 a Using KS test $\alpha = 0.05$, to test whether the data shown below are uniformly distributed.
0.44, 0.81, 0.14, 0.05, 0.93 08

b Using Chi - Square test with level of significance $\alpha = 0.05$, test whether the data shown below are uniformly distributed. With tests using 10 intervals of equal length and total number of sample data are 100.

Demand	1	2	3	4	5	6	7	8	9	10
Observations	8	8	10	9	12	8	10	14	10	11

08

Discrete Event System Simulation

V Semester SEE Question Papers

Course codes: 21IM55B1

- Apr/May 2024 (21IM55B1)

21IM55B1

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

V Semester B. E. Examinations Apr/May-2024

Industrial Engineering and Management

DISCRETE EVENT SYSTEM SIMULATION (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of statistical table is permitted.

PART-A

			M	BT	CO
1	1.1	Termination of a simulation depends upon _____.	01	1	1
	1.2	Time to failure for a component is modeled by _____.	01	2	1
	1.3	_____ distribution models different process when only the minimum, most likely and maximum values of the distribution are known.	01	2	4
	1.4	_____ is the first step in input modeling.	01	1	4
	1.5	Auto correlation tests the _____of the random numbers.	01	1	3
	1.6	Two important statistical properties of random numbers are _____ and _____.	02	2	2
	1.7	Frequency tests uses _____ and _____ tests.	02	2	2
	1.8	In the formula $X_i + 1 = (aX_i + c) mod\ m$ for linear congruential methods X_0 is called _____.	01	1	2
	1.9	Acceptance – Rejection technique is used to generate_____.	01	2	3
	1.10	A continuous system is one in which the state variables change only at a _____ set of points in time.	01	2	3
	1.11	A simulation system that runs continuously for long period of time is called _____.	01	1	2
	1.12	Kolmogorov-Smirnov test is used when sample size is _____ than 40.	01	2	2
	1.13	Simulation models that contain no random variable are classified as _____.	01	2	3
	1.14	Discrete moles are used to simulate only _____ events.	01	1	3
	1.15	Simulation of bank would involve random _____ and random service time.	01	1	4
	1.16	Exogenous describes _____ outside the system.	01	2	3
	1.17	Instantaneous occurrence that may change the state of the system is called as _____.	01	1	2
	1.18	Arena is written in _____ simulation language.	01	1	4

PART-B

2	a	Explain the steps involved in a simulation study with the help of a flow diagram.	08	2	1
	b	Name entities, attributes, activities, events and state variables for the following systems i) A Hospital emergency room. ii) A fast food restaurant.	08	3	1

3	a	Based on the runs up and down test determine whether the following sequence is such that the hypothesis of independence with $\alpha = 0.05$. 0.41, 0.68, 0.89, 0.94, 0.74, 0.91, 0.55, 0.62, 0.36, 0.27, 0.19, 0.72, 0.75, 0.08, 0.54, 0.02, 0.01, 0.36, 0.16, 0.28, 0.18, 0.01, 0.62, 0.73, 0.04, 0.83, 0.45, 0.13, 0.57, 0.63, 0.29	08	3	3
	b	Consider the 30 numbers given in the table. Test whether the 2nd, 6th, 10th and so on, numbers in the sequence are auto correlated, where $\alpha = 0.05$. 0.97 0.87 0.63 0.66 0.3 0.69 0.37 0.01 0.66 0.51 0.92 0.36 0.47 0.80 0.94 0.31 0.07 0.09 0.19 0.29 0.94 0.87 0.85 0.66 0.78 0.94 0.56 0.67 0.07 0.43	08	3	3
		OR			
4	a	A sequence of random numbers $0.44, 0.81, 0.14, 0.05, 0.93$ has been generated. Use Kolmogorov Smirnov test with a level of significance 0.10 to determine if the hypothesis that the numbers are uniformly distributed on the interval (0,1) can be rejected.	08	4	2
	b	Down times for a high production candy making machine have been found to be gamma distributed with mean 2.2 minutes and variance $2.10\,\text{min}^2$. Generate a gamma distributed random variate using acceptance – rejection techniques. Use the following random numbers :$0.83, 0.02, 0.43, 0.71$.	08	4	2
5	a	Explain the steps by step procedure for random variate generation using inverse transform technique.	08	2	3
	b	Explain briefly meaning of optimization in a simulation.	08	2	4
		OR			
6	a	Develop a random variate generator for triangular distribution with range (2,6) and mode at 3. Generate four values of random variates. Use $R = 0.36, 0.18, 0.89, 0.60$.	08	5	3
	b	Generate two random variates each from Weibull distribution (with scale & shape parameters as 10 and 2 respectively).	08	4	3
7	a	List and explain the types of simulation with reference to output analysis.	08	2	4
	b	Explain briefly point estimate and interval estimate as measures of performance.	08	2	3
		OR			
8	a	Explain terminating and steady state simulation with respect to output analysis.	08	2	2
	b	Briefly explain with examples the initialization bias in steady state simulation.	08	2	2
9	a	Explain the Finger three step approach to aid in validation process.	08	2	4
	b	Explain factors that are relevant in selection of simulation software.	08	3	4
		OR			
10	a	Explain the trends in simulation software.	08	2	2
	b	List the various simulation packages. Explain any three of them.	08	2	2

Advanced Manufacturing Processes

V Semester SEE Question Papers

Course codes: 16IM5A1

- Fast Track Oct 2020 (16IM5A1)
- Nov/Dec 2019 (16IM5A1)
- Nov/Dec 2018 (16IM5A1)

16IM5A1

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B. E. Fast Track Examinations Oct-2020

Industrial Engineering and Management

ADVANCED MANUFACTURING PROCESSES (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	In Electric discharge machining, the temperature developed is of the order of _____.	01
	1.2	In Electron bean machining, work piece is held in _____.	01
	1.3	Which method is used for profile cutting of flat plates?	01
	1.4	The process of utilizing mainly thermal energy for removing material is _____.	01
	1.5	In electrochemical machining (ECM) removal of metal from the work piece takes place _____.	01
	1.6	The process based on Faradays law of Electrolysis is _____.	01
	1.7	The vibrating frequency used for the tool in Ultrasonic machining is of the order of _____.	01
	1.8	Abrasive water jet cutting and abrasive jet machining are two different names for the same process :	
	1.9	In electron beam machining, as the electrons strikes the work piece _____ converted to _____.	02
	1.10	Plasma is a mixture of _____, _____ and _____.	02
	1.11	What are used as gas lasers in Laser beam machining?	02
	1.12	In which of the processes, a nozzle is used?	02
	1.13	In Abrasive jet machining which gas is not used?	02
	1.14	In Ultrasonic machining, the function of transducer is to _____.	02

PART-B

2	a	With the help of a neat diagram explain the mechanism of material removal in ultrasonic machining.	08
	b	Discuss about the major process parameters involved in Abrasive Jet Machining.	08
3	a	With the help of a neat diagram explain the principle of operation of EDM process.	
	b	Discuss about the EBM process parameters.	
		OR	

4	a	With the help of neat sketches explain what is meant by transferred and non-transferred mode of plasma arc. Mention the limitations of PAM.	08
	b	Discuss about the types of Lasers, advantages and limitations of Laser beam machining.	08
5	a	With the help of neat diagram explain the basic components of ECM process. Also mention the advantages and dis advantages.	08
	b	What is chemical machining? With the help of a neat diagram explain the steps to be followed in chemical machining.	08
		OR	
6	a	Discuss about the factors that affect the material removal rate in ECM.	08
	b	With the help of a neat diagram explain the working of ECG and also mention the advantages.	
7	a	With the help of a neat sketch explain Electro-hydraulic forming process. Also mention advantages and limitations of the process.	08
	b	Enumerate the need and applications of micromachining.	04
	c	Write a note on Laser micromachining.	04
8	a	What is $MEMS$? With the help of a schematic diagram discuss about the various components of $MEMS$. Also mention its advantages.	08
	b	Discuss briefly about the properties of carbon nanotubes.	04
	c	What are nanofabrication methods? Explain briefly.	04

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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-19

Industrial Engineering and Management

ADVANCED MANUFACTURING PROCESSES (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	The non-conventional machining process is also called as _____.	01
	1.2	The high tool consumption takes place in _____ process.	01
	1.3	In EDM the MRR is based on _____ principles.	01
	1.4	The word $LASER$ stands for _____ .	01
	1.5	Butyl is a _____.	01
	1.6	_____ fluids are used in electric discharge machining process.	01
	1.7	The frequency of above $16kz/s$ is used in _____ process.	01
	1.8	In ultrasonic machining, magnetostrictor converts magnetic energy into _____ energy.	01
	1.9	AJM process is recommended for _____ material.	01
	1.10	_____ Mechanical Machining Process requires low capital investment.	01
	1.11	List the different types of concentrators.	02
	1.12	Identify the parameters that affect the material removal rate in ECM.	02
	1.13	List the various etchants used in chemical machining.	02
	1.14	State the basic principle of high velocity forming process.	02
	1.15	What is meant by micro machining?	02

PART-B

2	a	How does the following parameters affect material removal rate (MRR) in abrasive jet machining i) Nozzle gap distance ii) Abrasive size iii) Nozzle pressure iv) Work material.	08
	b	With a schematic diagram, explain water jet machining process.	08
3	a	Explain with the help of neat sketches, the mechanism of material removal in EDM process. State its advantages and disadvantages.	10
	b	List some of the di-electric fluids used in EDM. What are the basic requirements of the di-electric fluids in EDM process?	06

		OR	
4	a	Explain the principle and operation of LBM process. Mention some advantages and limitations of LBM.	10
	b	List the applications of plasma arc machining process.	06
5	a	With a schematic diagram explain the electrochemical grinding process.	08
	b	Calculate the MRR and electrode feed rate in an ECM of iron (Fe) that has a cross sectional area of $30 \times 30\, mm$ with $NaCl$ in water as electrolyte. The gap between the tool and workpiece is $0.3\, mm$. The supply voltage is 12 volts DC & specific resistance of electrolyte is $3\, ohm - cm$. Given the atomic weight of Iron = 55.85, valence 2, density $7.87 \times 10^6 gm/m^3$.	08
		OR	
6	a	With the help of neat sketch explain the different steps involved in chemical machining process.	10
	b	Explain in brief, Electrochemical Deburring process.	06
7	a	Explain in detail, explosive forming process, with the help of a sketch.	08
	b	What is meant by micro-machining? Explain ultrasonic micro machining process.	08
8	a	Discuss the application of MicroElectroMechanical systems.	08
	b	Provide a brief note on nanofabrication methods.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-18

Industrial Engineering and Management

ADVANCED MANUFACTURING PROCESSES (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

PART-A

1	1.1	In Ultra sonic Machining, magnetostrictor converts magnetic energy into _____ energy.	01
	1.2	At what rate slurry is pumped through nozzle in USM?	01
	1.3	The Electrical Discharge machining (EDM) process is _____ free.	01
	1.4	The cathode filament is heated to a temperature of _____ ℃ in case of electron beam machining.	01
	1.5	Time of machining in laser beam machining process increases with _____ in laser beam spot diameter.	01
	1.6	The grinding wheel used in the Electrochemical grinding process is given _____ charge.	01
	1.7	What happens to the over-cut size, if we use the electrolyte flow mode under backward pressure?	01
	1.8	The system which consists of electrolytic solution and electrodes can be referred as _____ cell.	01
	1.9	The value of the amplitude in USM obtained without mechanical amplifier is from _____ to _____.	02
	1.10	In Electrical Discharge Machining(EDM), the spark gap is kept between _____ mm to _____ mm.	02
	1.11	What is the value of voltage that the power supply unit supplies for ECM?	02
	1.12	List the components for which, Electrochemical Deburring is applied?	02
	1.13	List the operations for which the Explosive forming process is used.	02
	1.14	List the parameters that are increased by using Micro machining processes.	02

PART-B

2	a	List and explain the aspects that are considered in process selection of advanced manufacturing methods.	06
	b	With a neat sketch explain the components of water jet machining.	10

3	a	Enumerate the principle of an electric discharge machine with a neat sketch.	10
	b	List and explain the factors that determine the suitability of electrode material in electric discharge machine.	06
		OR	
4	a	Enumerate the working principle of plasma arc machining process with a neat sketch.	10
	b	List at least three advantages and disadvantages of plasma arc machining process.	06
5	a	Classify and differentiate between the methods for depositing the maskant on to the work piece surface in chemical manufacturing processes.	06
	b	Explain the process characteristics of electro-chemical grinding.	10
		OR	
6	a	In a electro chemical dissolution process of iron, a metal removal rate of $2cc/min$ was desired. Determine the amount of current required for the process, assuming atomic weight of iron=$56gm$, valence at which dissolution $= 2, F = 1609\ amp.min$ and density of iron= $7gm/cc$.	10
	b	State the practical applications of electro chemical machining.	06
7	a	State the principle and discuss the process parameters of Electro Hydraulic Forming.	10
	b	Analyze the following factors, based on the constant feed of power in abrasive micro machining: i) Effect of reservoir powder level ii) Powder stratification iii) Powder humidity.	06
8	a	List and explain the additive films and materials from micro electro mechanical systems.	10
	b	Give examples for the following process types: i) Thin film deposition ii) Etching iii) Machining.	06

Modern Manufacturing Methods

V Semester SEE Question Papers

Course codes: 12IM5A1

- Fast Track Jul 2018 (12IM5A1)
- Nov/Dec 2017 (12IM5A1)
- Nov/Dec 2016 (12IM5A1)

12IM5A1

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Fast Track July - 18

Industrial Engineering and Management

MODERN MANUFACTURING METHODS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

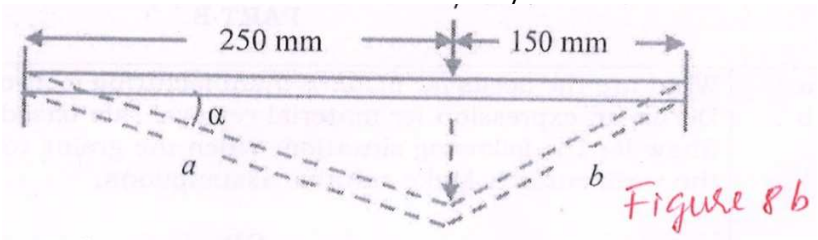
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	What are the factors considered while selecting modern manufacturing process based on physical parameter?	02
	1.2	Why tool feed mechanism is needed in Ultrasonic machining process?	02
	1.3	What are the reactions taking place at anode and cathode for dissolution of iron in electrochemical machining. Assume electrolyte as water.	02
	1.4	In a certain electrochemical dissolution process of iron a metal removal rate of $2\,cm^3/min$ was desired. Determine the amount of current required for the process if atomic weight $= 56\,g$, valvancy $= 2$ and density $= 7.8\,g/cm^3$.	02
	1.5	What are the functions of dielectric fluid used in electric discharge machining process?	02
	1.6	What are the different methods of flushing?	02
	1.7	Estimate the force required in punching a $50\,mm$ diameter hole through a $5\,mm$ thick annealed aluminum alloy $6A1-4V$ sheet at room temperature. Take ultimate strength of alloy is $1000\,MPa$.	02
	1.8	A $10\,mm$ steel sheet is bent to radius of $105\,mm$. Assuming that its yield stress, $210\,MPa$ and modulus of elasticity is $200\,GPa$. Calculate the radius of the part after bent.	02
	1.9	Why clean room is required in manufacturing of integrated circuit board?	02
	1.10	List four types of lithography techniques.	02

PART-B

2	a	What are the needs for modern manufacturing methods?	04
	b	Derive an expression for material removal rate based on the theory of Shaw for the following situation: When the grains to be thrown on to the work surface. Make suitable assumptions.	12
		OR	

3	a	With a neat sketch explain the construction and working principle of Ultrasonic machining process.	08
	b	Explain briefly the parameters influencing the material removal rate in abrasive jet machining process.	08
4	a	Derive a theoretical relationship for the determination of the metal removal rate in electrochemical machining. When the work place is of pure metal and alloy.	10
	b	Calculate the machining rate and electrode feed rate when iron is electrochemically machined, using copper electrode and sodium chloride solution (specific resistance = 5.0Ωcm). Assume Atomic weight of iron: 56 valance of iron: 2 density of iron: $7.87 \times 10^6\ (g/m^3)$ Supply voltage: $18\ V\ DC$ Current: $5000\ amp$ Inter electrode gap: $0.5\ mm$	06
		OR	
5	a	With a schematic diagram explain the chemistry of the electro chemical machining process.	08
	b	Explain the various steps involved in chemical milling process with a block diagram.	08
6	a	Briefly explain mechanism of metal removal in electric discharge machining process.	06
	b	What is flushing? With a neat sketch explain any two types of flushing used in EDM process.	10
		OR	
7	a	With a neat sketch explain working principle of traveling wire EDM.	08
	b	Explain with a neat sketch the generation of plasma arc in PAM and give the various applications of PAM.	08
8	a	What are general characteristic that you are going to consider in manufacturing of outer body of a CAR?	08
	b	A $400\ mm$ long sheet with a cross sectional area of $300\ mm^2$ (figure 8b) is stretched with a force F, until $\alpha = 20°$. The **material** has a true – stress – true – strain curve $\sigma = 700 \times 10^6 \boldsymbol{N/mm^2}$  Figure 8b i) Find the total work done, ignoring end effects and bending. ii) What is α_{max} before necking begins?	08

		OR	
9	a	Consider the case of a coffee mug being produced by rapid prototyping. Describe how the top of the handle can be manufactured, since there is no material directly beneath the arch of the handle.	08
	b	With a schematic diagram explain the working principle of stereolithography.	08
10	a	Explain any two film deposition technique used in fabrication of microelectronic, micromechanical, microelectromechanical devices.	08
	b	What is the term oxidation in MEMS? Explain three oxidation techniques used in modern IC technology.	08
		OR	
11	a	With a schematic diagrams explain surface micromachining.	10
	b	What is the difference between isotropic and an isotropic etching?	06

12IM5A1

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. ExaminationsNov/Dec-17

Industrial Engineering and Management

MODERN MANUFACTURING METHODS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

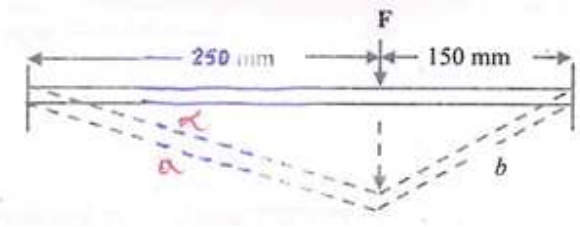
1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Modern manufacturing methods are classified based on _____ utilization.	01
	1.2	Tool cone in USM is also called as _____.	01
	1.3	Explain the functions of tool feed mechanism used in ultrasonic machining process.	02
	1.4	Name the chemical reactions taking place at anode and cathode for dissolution of iron in electrochemical machining. Assume electrolyte as water.	02
	1.5	While machining tungsten a metal removal rate of $0.98\,cm^3/min$ has been observed at $1000\,A$. Evaluate the valence at which the metal has been electrochemically dissoluted if Atomic weight $186g$ and density $= 19.4\,g/cc$.	02
	1.6	Why dielectric fluid is used in electric discharge machining process?	02
	1.7	List any four methods of flushing used in electric discharge machining process?	02
	1.8	Estimate the force required in punching a $50\,mm$ diameter hole through a $3\,mm$ thick annealed aluminum alloy $6A1-4V$ sheet at room temperature. Take ultimate strength of alloy as $1000\,MPa$.	02
	1.9	A $10\,mm$ steel sheet is bent to radius of $105\,mm$. Assuming that its yield stress is $210\,MPa$ and modulus of elasticity is $200\,GPa$. Calculate the radius of the part after bent.	02
	1.10	The process by which the geometric patterns that define devices are transformed from a photomask to the substrate surface is called as _____.	01
	1.11	The acronym of $MEMS$ in advanced manufacturing method is _____.	01
	1.12	List four types of lithography techniques.	02

PART-B

2	a	In what way modern machining differs from conventional machining process.	06
	b	Derive an expression for material removal rate based on the theory of Shaw for the following situation: when the grains to be thrown on to the work surface. Make suitable assumptions.	10
		OR	
3	a	With a neat sketch explain the construction and working principle of Abrasive jet machining process.	08
	b	Explain briefly the parameters influencing the material removal rate in ultrasonic machining process.	08
4	a	Derive a theoretical relationship for the determination of the metal removal rate in electrochemical machining, when the work piece is of pure metal and alloy.	10
	b	Calculate the machining rate and electrode feed rate when iron is electrochemically machined, using copper electrode and sodium chloride solution (specific resistance = 5.0Ω). Assume $Atomic\ weight\ of\ iron$: 56 $Valancy\ of\ iron$: 02 $Density\ of\ iron$: $7.87 \times 10^6\ g/m^3$ $Supply\ voltage$: $18V\ DC$ $Current$: $5000A$ $Inter-electrode\ gap$: $0.5\ mm$	06
		OR	
5	a	With a schematic diagram explain the working principle of electro chemical grinding process.	08
	b	With a schematic block diagram explain the various steps involved in chemical blanking process.	08
6	a	With a neat sketch explain pressure flushing and side flushing used in EDM process.	10
	b	With a neat sketch explain working principle of traveling wire EDM process.	06
		OR	
7	a	Explain with a neat sketch the generation of plasma arc in PAM and give the various applications of PAM.	10
	b	Explain what is meant by transferred and non –transferred mode of plasma torch.	06
8	a	Briefly explain general characteristics that are considered in sheet metal forming process.	08

	b	A 400 mm long sheet with a cross sectional area of 300 mm^2 is sketched with a force F, until $\alpha = 20°$. The material has a ture-stress-true-strain curve $\sigma = 700 \times 10^6 \epsilon^{0.3}$  i) Find the total work done, ignoring end effects and bending ii) What is α_{max} before necking begins?	08
		OR	
9	a	With a schematic diagram explain the working principle of fused deposition modeling process.	08
	b	With a schematic block diagram explain laser sintering process.	08
10	a	List and explain any two film deposition techniques used in fabrication of micro-electromechanical devices.	08
	b	What is the term oxidation in $MEMS$? Explain three oxidation techniques used in modern IC technology.	08
		OR	
11	a	With an appropriate sketch describe the thermo sonic stitching process.	10
	b	What is the difference between isotropic and an-isotropic etching?	06

12IM5A1

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

MODERN MANUFACTURING METHODS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	In *USM* process, the material is by the _____.	01
	1.2	Modern manufacturing methods are classified based on _____ utilization.	01
	1.3	The *USM* process has about four types of material removal phenomenon involved namely _____.	02
	1.4	In abrasive jet machining the most favorable grain size ranges from _____.	01
	1.5	The friction welding is a _____ state welding process that produces a weld under _____ forces.	02
	1.6	The principal advantage of *ECM* is _____.	01
	1.7	In deep drawing the depth of the part being made is more than _____ of its diameter.	01
	1.8	The two processes used for removing very fine burrs of metal that are left on the work piece are _____ and _____.	02
	1.9	_____ process builds plastic parts or objects , a layer at a time by tracing a laser beam on the surface of a vat of liquid photo polymer.	01
	1.10	With is meant by Thin films.	02
	1.11	Electronic semi-conductor devices and optical coatings are the applications of _____ construction.	01
	1.12	Expansion of *HEPA* is _____.	01
	1.13	Define lithography.	02
	1.14	List any four operations that are based on the shearing process.	02

PART-B

2	a	List the advantages and applications of ultrasonic machining.	08
	b	With a neat sketch explain the working principle of Abrasive Jet Machining.	08
		OR	
3	a	List the applications of friction welding.	06
	b	With neat sketch explain the process of Orbital riveting and forming.	10

4	a	During electrochemical machining of pure iron a material removal rate of $600mm^3/min$ is required. Estimate current requirement for the same.	06
	b	Explain the process of electrochemical grinding with the help of a sketch.	10
		OR	
5	a	Explain the method of masking and etching in chemical blanking process.	08
	b	List the advantages and disadvantages of Chemical Milling process.	08
6	a	Explain different types of electrode materials used in EDM process.	06
	b	Explain plasma arc machining process with a neat sketch.	10
		OR	
7	a	Explain the working of wire cut EDM process with a sketch.	10
	b	State the process parameters of plasma arc machining.	06
8	a	Explain the various sheet metal forming process with their important characteristics.	10
	b	Differentiate between SLA and SLS in Rapid prototyping process.	06
		OR	
9	a	What is meant by shearing? List the characteristics of shearing process. Supplement your answer with an illustration of shearing process.	10
	b	Analyze FDM process in terms of its advantages and disadvantages.	06
10	a	Evaluate suitability of the various techniques involved in film deposition for a variety of pressures, temperatures and vacuum systems.	10
	b	Explain the process involved in the manufacturing of hearing aids.	06
		OR	
11	a	Evaluate the process of Ion implantation with an example.	08
	b	Elaborate on the principles of clean room testing.	08

Advanced Ergonomics

V Semester SEE Question Papers

Course codes: 12IM5A2

- Nov/Dec 2017 (12IM5A2)
- Nov/Dec 2016 (12IM5A2)
- Fast Track Jul 2016 (12IM5A2)

12IM5A2

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

ADVANCED ERGONOMICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	The first objective of human factors engineering is to enhance the _____ and _____ with which the work and other activities are carried out.	01
	1.2	List the fundamental characteristics of systems.	02
	1.3	The three main classes of human-machine system are _____, _____ and _____.	02
	1.4	System reliability is usually expressed as the probability of successful performance. Another measure of reliability is _____.	01
	1.5	The three common types of quantitative displays are _____, _____ and _____.	02
	1.6	_____ refers to the ability of the lens of the eye to focus the light rays on the retina.	01
	1.7	_____ is expressed as the ratio of the thickness of the stroke to the height of the letter or number. While the relationship between the width and height of a complete alphanumeric character is describes as the _____.	02
	1.8	The amount of energy needed per unit time to sustain life is called the _____.	01
	1.9	The factors that affect the level of energy consumption on a particular task are _____, _____ and _____.	02
	1.10	The fundamental photometric quantity is luminous flux. The unit of luminous flux is _____.	01
	1.11	Relative humidity is measured using a _____.	01
	1.12	The instrument used to measure a person's threshold of hearing at selected frequencies is called as _____.	01
	1.13	_____ is the rate of change of motion of an object having some mass.	01
	1.14	_____ is an "inappropriate or undesirable human decision or behavior that reduces, or has the potential for reducing, effectiveness, safety or system performance".	01
	1.15	A condition or set of circumstances that has the potential of causing or contributing to injury or death is called as _____.	01

PART-B

2	a	Explain the basic functions performed by human or machine components of human-machine system.	10
	b	Explain the fundamental characteristics of human-machine systems.	06
		OR	
3	a	Explain with the help of schematic diagram the various components of human-machine system.	10
	b	Explain in detail the steps involved in conducting rapid upper limb assessment.	06
4	a	Explain the factors affecting the visual acuity and contrast sensitivity.	08
	b	Explain the factors influencing the detectability of signal and warning lights.	08
		OR	
5	a	Discuss with the help of suitable examples perceptual principles of symbolic design.	10
	b	Explain the three applications of tactual displays as substitute for hearing.	06
6	a	Explain with the help of a block diagram the factors influencing the power and capacity for physical activity.	10
	b	List various ways of reducing the risk of manual materials in handling overexertion.	06
		OR	
7	a	Define strength. Explain the various measures of strength and factors affecting strength.	06
	b	Explain the three distinct approaches for assessing manual material handling capabilities.	10
8	a	Compare the different avenues of heat exchange between the body and the environment.	08
	b	Discuss the various antireflection techniques available for visual display terminals.	08
		OR	
9	a	Discuss the various means of controlling noise.	08
	b	Explain the effects of vibration on the performance of human operator subjected to vibration.	08
10	a	Enumerate the major stages in the system design process.	08
	b	Explain with the help of a block diagram various stages in occurrence or avoidance of accidents in a potentially hazards situation.	08
		OR	
11	a	Elucidate discrete-action human error classification schemes.	08
	b	Discuss the various theories of accident causation.	08

12IM5A2

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

ADVANCED ERGONOMICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	What are the characteristics of a system?	02
	1.2	Quote an example for reliability of a system.	02
	1.3	What is redundancy? How to equate percentage of redundancy?	02
	1.4	Human failures are classified as _____ and _____.	02
	1.5	What are the steps involved in checking product liability?	02
	1.6	What are the two ways of identifying signals?	02
	1.7	What is the difference between alarm and signal?	02
	1.8	For reasonably fit adult male, the "grade of physical work" is based on _____.	01
	1.9	List four types of human functions or tasks involved in reception of auditory signals.	02
	1.10	Distinguish between hazards and risk.	02
	1.11	Mention any one Risk assessment tool used in tracking musculo skeletal disorders ($MSDs$)	01

PART-B

2	a	Explain in detail the procedure involved in conducting assessment of Musculo Skeletal Disorders ($MSDs$) and cumulative trauma disorders.	10
	b	Illustrate the dimensions of coding of information.	06
		OR	
3	a	Explain the types of Compatibility in the field of ergonomics. Support your answer with suitable examples.	08
	b	With the help of graphical representation, explain the outcomes of decision making when there is an interference of noise in information over a period of time.	08

4	a	What is work efficiency? How it can be measured? Illustrate factors influencing power and capability for physical activity.	10
	b	Discuss different approaches of assessing manual material handling (MMH) capabilities. Quote suitable examples.	06
		OR	
5	a	What is sound localization? Explain in detail with suitable examples.	08
	b	Analyze the factors influencing tracking performance in human control system.	08
6	a	List and explain the characteristics of auditory displays.	06
	b	How does the physiological strain from physical work is measured through macro level and physiological responses?	10
		OR	
7	a	How the work out at a gymnasium is different from a heavy work out at shop floor (MMH)? Support your answer with examples.	08
	b	Illustrate the stress and strain relation concern with human beings at work stations.	08
8	a	What is glare? Discuss the techniques of reducing the reflection under VDT.	08
	b	Discuss the effects of lighting on performance, considering the experiment conducted at Hawthorne as the case.	08
		OR	
9		Generate a detailed report on "Study on comparative analysis of telecast of cricket match on television in present and previous years". Highlight the advancements of $VDTs$ that facilitated in ease of information transmission.	16
10	a	How the human failures are dealt to reduce the likelihood of and consequences of human errors?	08
	b	"Accidents can be reduced by altering the behaviors". Examine the statement in the context of vehicle drivers.	08
		OR	
11		Considering a live example of your choice as a system, discuss in detail all the stages to be satisfied under system design process. Represent the discussion with a block diagram.	16

12IM5A2

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Fast Track Examinations July-16

Industrial Engineering and Management

ADVANCED ERGONOMICS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	Mention the two objectives of human factors.	02
	1.2	What is meant by the term 'System'?	02
	1.3	What is the unit of measure of information?	02
	1.4	What is meant by Compatibility?	02
	1.5	What is maximum Aerobic Power?	02
	1.6	What is the expanded form of acronym *NIOSH*?	02
	1.7	List the characteristics of lifting object that influence biomechanical, physiological and psychophysical stresses.	02
	1.8	Mention the environmental situations that lead to draught effect.	02
	1.9	Mention the symptoms of motion sickness.	02
	1.10	List the four principal purposes of warning.	02

PART-B

2	a	Give a brief overview of major human factors developments.	08
	b	With the aid of a schematic representation, explain various components of a human machine system.	08
		OR	
3	a	Discuss briefly the fundamental characteristics of systems, with special reference to human-machine systems.	12
	b	Write a short note on System Reliability.	04
4	a	Provide a detailed classification of Information Presented by displays.	08
	b	Bring out the relationship between visual acuity and accommodation of eyes.	08
		OR	

5	a	List and explain the factors affecting visual acuity and contrast sensitivity.	08
	b	List any eight circumstances in which auditory displays would be preferable to visual displays.	08
6	a	Elaborate on the factors that affect the level of energy consumption on a particular task.	08
	b	Discuss in brief, the three approaches for assessing manual material handling capabilities.	08
		OR	
7	a	Explain the relevance of job design, work selection and work training as tools for reducing the risk of modern material handling over exertion.	12
	b	List the factors influencing tracking performance.	04
8	a	What is meant by Heat Stress? Elaborate on the factors that result in considerable difference between people with respect to their tolerance to heat stress.	08
	b	What is meant by cold stress? Explain various means of protecting form cold stress.	08
		OR	
9	a	List the acoustic and non-acoustic factors that influence the annoyance quality of noise.	08
	b	Explain the effects of vibration on performance.	08
10	a	Explain the human error classification scheme proposed by Rouse and Rouse.	08
	b	Provide an overview of three categories of Accident Causation.	08
		OR	
11	a	Discuss the major stages in system design process.	12
	b	List the personal characteristics that can affect driver's driving performance.	04

Database Management Systems

V Semester SEE Question Papers

Course codes: 12IM5B5

- Nov/Dec 2017 (12IM5B5)
- Nov/Dec 2016 (12IM5B5)

12IM5B5

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

DATABASE MANAGEMENT SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

11. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
12. Answer FIVE full questions from Part B.

PART-A

1	1.1	What is data base and database management system?	02
	1.2	List the advantages of database management system.	02
	1.3	What are the three basic notions that the $E-R$ data model?	02
	1.4	What are the basic issues in the design of an $E-R$ database schema?	02
	1.5	Distinguish between single valued and multivalued attributes.	02
	1.6	What are data base languages?	02
	1.7	Differentiate between LAN and WAN.	02
	1.8	How the data is stored on disc?	02
	1.9	What is inheritance in object oriented databases?	02
	1.10	List any four Relational databases new applications that have been in commercial world.	02

PART-B

2	a	A major purpose of a database system is to provide users with an abstract view of data. Justify the statement how abstract view of data can be done?	08
	b	Explain with neat sketch the simplified data base system environment.	08
		OR	
3	a	Explain briefly with neat sketch the tree schema architecture of $DBMS$.	08
	b	Explain with neat sketch the two tier client/server architecture of $DBMS$.	08
4	a	List and Name the notations used for $E\,R$ Diagram.	08
	b	What is data model? Explain different types of data models with neat sketches.	08
		OR	
5	a	Discuss the different types of mapping constraints.	08

	b	A car has certain attributes year, registration, vehicle_id, model, color, make. Registration Attribute is further subdivided into two simple attributes and cars are available in two colors red and blue. i) Identify entity, key attribute, composite attribute, multi-valued attributes ii) Using $E-R$ diagram notations draw ER diagram iii) Draw entity set with two entities(assume some attribute values on your own)	08
6	a	List and illustrate all relational algebra operations of set theory.	08
	b	Explain briefly the entity integrity constraint, referential integrity constraint and foreign keys with respect to relational database.	08
		OR	
7	a	Discuss the differences between $DBMS$ and $RDBMS$.	04
	b	What are the Codd's twelve rules in $RDBMS$.	12
8	a	List and explain schema change statement in SQL.	08
	b	Briefly illustrate the three modify SQL commands used for modifying the database.	08
		OR	
9	a	List and explain the four informal design guidelines for developing relation schemas.	08
	b	What is Normal Form? Illustrate all the three forms of relations.	08
10		What is File Organizations? Discuss the different Types of File Organization.	16
		OR	
11	a	What is buffering of blocks? With neat sketch explain the process of buffering of blocks.	08
	b	What is indexing? Explain different types of indexing.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

DATABASE MANAGEMENT SYSTEMS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

10. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
11. Answer FIVE full questions from Part B.

PART-A

1	1.1	Define the concept of data independence in database management system.	02
	1.2	What is the difference between a database scheme and a database state?	02
	1.3	List the two cases where use of a $NULL$ value would be appropriate.	02
	1.4	Distinguish between single valued and multivalued attributes.	02
	1.5	Mention the three main categories of constraints on databases.	02
	1.6	Why entity integrity and referential integrity constraints are considered to be important?	02
	1.7	List the data types that are allowed for SQL attributes.	02
	1.8	Define $BCNF$.	02
	1.9	How does double buffering improve block access time?	02
	1.10	Differentiate between fixed length records and variable length records.	02

PART-B

2	a	If you were designing a web based system to make airline reservations and sell airline tickets, which $DBMS$ architecture would you choose? Why? Why would the other architecture not be a good choice.	08
	b	Examine critically how three scheme architecture enables the database approach to separate user applications and physical database. Support your answer with an illustration.	08
		OR	
3	a	Categorize different types of database end users based on the activities performed.	08
	b	State five main advantages of using $DBMS$.	08

4 a Suppose you are given the following requirements for a simple database for the National Hockey League (*NHL*):

- The *NHL* has many teams
- Each team has a name, a city, a coach, a captain and a set of players
- Each player has a name, a position (such or left wing as goalie), a skill level and a set of injury records
- A team captain is also a player
- A game is played between two teams (returned to as host-team and guest team) and has a date (such as May 11th, 1999) and a score (such as 4 to 2)

Construct a clean and concise *ER* diagram for *NHL* database. 10

b What is meant by relationship in an *ER* diagram? Provide examples for three types of binary relationship that exists amongst entities. 06

OR

5 a A university registrar's office maintains data about the following entities:

- Courses including numbers, title, credits, syllabus and prerequisites
- Course offerings, including course number, year, semester, section number, instructors, timings and class room
- Students, including student id, name and program
- Instructors, including identification number, name, department and title.

Draw the *ER* diagram. 10

b Explain the differences among an entity, an entity type and entity set. 06

6 One possible database state for the company national database scheme.

Employee

First name	*initials*	*Last name*	*SSN*	*Birth date*	*Address*	*Sex*	*Salary*	*Super SSn*	*D. No.*
John	*B*	*Smith*	123456789	1965-01-09	*Fondren*	*M*	30000	333445555	5
Franklin	*T*	*Wory*	333445555	1955-12-08	*Vugs*	*M*	40000	888665555	5
Alicia	*J*	*Zelaya*	999887777	1968-01-19	*Castle*	*F*	25000	987654321	4
Jennifer	*S*	*Wallace*	987654321	1941-06-20	*Berry*	*F*	43000	888665555	4
Ramesh	*K*	*Narayan*	666884444	1962-09-20	*Fire oak*	*M*	38000	333445555	5
Joyce	*A*	*English*	453453453	1972-07-31	*Rice*	*F*	25000	333445555	5
Ahmad	*V*	*Jabbas*	987987987	1969-03-29	*Dallas*	*M*	25000	986754321	4
James	*E*	*Borg*	888665555	1937-11-10	*Stone*	*M*	55000	*NULL*	1

Department

DName	*DNo.*	$Ngr - SSn$	$Mgr - start\ date$
Research	5	333445555	$1988 - 05 - 22$
Admin	4	986754321	$1995 - 01 - 01$
Head	1	888665555	$1981 - 06 - 19$

Department locations

$Dnumber$	$Dlocation$
1	$Houston$
4	$Stafford$
5	$Bellaire$
5	$Sugarcane$
5	$Houstan$

Determine if any integrity constraints will be violated by each of the following operations

a) $insert\ into\ department\ values\ (\text{“}Audit\text{”}, 7, 998800000, null)$;

b) $insert\ into\ department\ values\ (\text{“}maintenance\text{”}, null, 4534553, 2013_02_01)$;

c) $update\ Employee\ set\ Super_{ssn} = 987654321\ where\ SSn = 333445555$;

d) $delete\ from\ department\ where\ mgr_ssn = 88866555$ — 16

OR

7 a Consider the following relations for a database that keeps track of business trips of salespersons in a sales office
$SALESPERSON\ (SSN, Name, Start_year, Dept.no)$
$TRIP\ (SSN, From_city, T_city, Dept_date, Return_date, Trip_ID)$
$EXPENSE\ (Trip_Id, Account\ \#, amount)$
Specify the foreign keys for this schema, stating any assumptions you make. — 06

b Explain critically the guidelines as advocated by EF codes for the design of relational database. — 10

8 a Write a SQL statement to create a table job_history including columns employee_id, start_date, end_date, job_id and department_id and make sure that, the employee_id column does not contain any duplicate value at the time of insertion and the foreign key column job_id contain only those values which exists in the jobs table. — 08

b SQL is both a data definition language and data manipulation language. Justify the statement with examples. — 08

OR

9 a Why is normalization necessary while designing the relational database? Analyze various levels in normalization process. — 08

b Examine the informal measures of quality for relational database design. — 08

10 a Provide an overview on the mechanism of reading data from and writing data to the disk. — 08

b Discuss the advantages of ordered files over unordered files. — 08

OR

11 a Multilevel indexing through B tree and B^+ tree improves the efficiency of searching an index file. Justify. — 10

b Examine the features of the three types of ordered single level indexes. — 06

Design of Experiments

V Semester SEE Question Papers

Course codes: 12IM5A3

- Nov/Dec 2017 (12IM5A3)
- Nov/Dec 2016 (12IM5A3)

12IM5A3

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

DESIGN OF EXPERIMENTS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B.
5. Use of orthogonal array tables and statistical tables is permitted.

PART-A

1	1.1	Creating a 9-level design using an orthogonal array with only 3-level columns is achieved using the technique of _____.	01
	1.2	If the fractions defective in a lot of products is 0.1, the corresponding S/N ratio is _____.	01
	1.3	The 2.5% point of a F-distribution with numerator and denominator degree of freedom 3 and 15 is _____.	01
	1.4	The interaction between col. 9 and col. 10 factors in the array $L_{27}3^{13}$ can be estimated from col. no. _____.	01
	1.5	When main factor – 2 way interaction is present in a fractional design, the resolution can not be _____.	01
	1.6	Count the degrees of freedom required to study factors A, B, C at 2 levels each, D & E at 3 levels each, and interactions AB, CD, DE.	01
	1.7	The 3 basic principles of experimentation are _____, _____, and blocking.	02
	1.8	A_0 and Δ_0 in quadratic loss functions denote _____ and _____ respectively.	02
	1.9	Interpret a P-value of less than significance level in an $ANOVA$ table.	02
	1.10	The total sum of squares for the set of observations $5, 6, 7, 8, 5, 6, 7, 8, 8, 9$ is _____.	02
	1.11	The response totals in two blocks of 4 runs each are 400×500. If only single replicates are involved, the block sum of squares is _____.	02
	1.12	The sources of noise can be classified as external, _____ and _____.	02
	1.13	The signal to noise ratio corresponding to the fill volume in a container, measured for 5 replications as $100, 80, 92, 102, 83$ is _____.	02

PART-B

2	a	Bring out the complimentary role of on-line and off-line quality control methods in quality improvement with specific reference to a manufacturing industry.	08
	b	Classify the noise factors and provide examples of noise factors for a refrigerator and an automobile for each of the classification.	08

OR

3 a A television set has an important characteristic x with functional limits in $50 \leq x \leq 90$, and a target of 70. The manufacturer has observed the cost to repair when x is at the limits to be $Rs.1{,}500$. Derive the corresponding quadratic loss function and draw the shape. 08

b With the above situation, what is the average loss of 4 sets with x measuring $58, 62, 70, 52$? 06

c By how much can the average loss be reduced if the deviations from target are decreased to 70% in each step? 02

4 Effective life of batteries were tested at various combinations of plate material(Type $1, 2$ or 3) and operating temperature($15, 70$ and $125°F$). Four batteries are tested at each combination, and the total are given as below(the square of 36 individual values sum upto 478,547)

	$Temp$		
Mtl	15	70	125
1	539	229	230
2	623	479	198
3	576	583	342

Conduct an $ANOVA$ to identify significant factors. 16

OR

5 The surface finish of cut wood is affected by type of wood, feed rate and depth of cut. Two replications are carried out for each of the combinations and the surface finish measurements are:

		$Depth\ of\ cut(mm)$			
		1		3	
		$Feed\ rate\ (m/mm)$		$Feed\ rate(m/mm)$	
		2	4	2	4
$Type\ of\ Wood$	Oak	$6, 8$	$18, 12$	$20, 26$	$36, 44$
	$Pine$	$10, 16$	$12, 9$	$34, 28$	$25, 22$

a) Determine all the main effects and interaction effects. 10

b) Draw the interaction plots for AB, AC and BC interactions. 06

6 How do you evaluate fractional designs in terms of their resolutions? Differentiate among resolution III, IV and V designs and provide examples for each design. 16

OR

7 A single replicate of four factor half-fraction experiment yielded the following responses:

$Treatment$	(1)	ad	bd	ab	cd	ac	bc	$abcd$
$Response$	45	100	45	65	75	60	80	96

a) Identify the defining relation and hence derive the full alias structure. 04

b) Comment on the estimated main and interaction effects. 08

c) Suggest an appropriate regression model($ANOVA$ is not required). 04

8 In a textile processing experiment, 4 factors A, B, C, D are to be studied, each at 3 levels:

	Factors	*Levels*
A	*Temperature of both* (℃)	$40, 48, 55$
B	*Voltage of heating* (V)	$3, 4, 5$
C	*Time of immersion* $(seconds)$	$30, 40, 50$
D	*Dye concentration in both* (%)	$30, 35, 40$

Derive the physical layout of the design to estimate all the main effects and interactions AB, BC, AC. 16

OR

9 Derive appropriate orthogonal array designs and provide design matrices.

a) Factor A at 2 levels, B, C and D at 3 levels each, no interactions 03

b) Factor A and E at 2 levels each, factors B, C and D at 3 levels each and no interactions. 03

c) Factors A, B, C and D at 2 levels each, and factors E at 4 levels, no interactions. 04

d) Factors A, B, C at 2 levels: depending on level of C, either the pair D and E, or the pair F and G has to be studied. No interactions are required. 06

10 Four replicates were run for a 2^2 experiment and the defect counts were recorded as below:

run	*Number of defects*
(1)	$6, 8, 15, 9$
a	$30, 18, 9, 11$
b	$10, 25, 3, 8$
ab	$25, 3, 24, 7$

Using the categories
I: $0 - 5$ defects
II: $6 - 20$ defects
III: 21 defects or more,
Analyze the data as ordered
Categorical data and determine the best level of factors to minimize defect count. 16

OR

11 a A communication subsystem is checked for transmission errors when signals are transmitted. The incidences of signal bit and transmitted bit are observed and found that the error of when 0 and 1 are transmitted are given as 10% and 5% respectively. What is the S/N ratio, considering this as a digital to digital dynamic system? 12

b Provide an example each for situations where static S/N ratios and dynamic S/N ratios are used in robust design. 04

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R. V. COLLEGE OF ENGINEERING

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V Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

DESIGN OF EXPERIMENTS (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

3. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
4. Answer FIVE full questions from Part B.
5. Use of statistical tables is permitted.

PART-A

1	1.1	Why is screening experiment performed?	01
	1.2	What are factor levels? How do you decide on the number of levels?	02
	1.3	What is an interaction?	01
	1.4	Define the main effect for a 2-level experiment.	01
	1.5	What is a Full Factorial experiment?	01
	1.6	What is a treatment combination? Explain with an example.	02
	1.7	How is aliasing useful in a fractional factorial experiment?	02
	1.8	What do the effects plots depict?	01
	1.9	Differentiate between fixed effects and random effects models.	02
	1.10	In factorial (2^k) design, factors are assumed to be _____ effects model.	01
	1.11	When do you say a product is robustly designed?	01
	1.12	The value of 'η' under the optimum condition is predicted using the additive model equation _____ (consider factors A and B with optimal level 1 each)	01
	1.13	State the stages of Taguchi's 3-stage design process.	02
	1.14	Draw the traditional model of quality loss function.	01
	1.15	Consider the following experiment. Determine the effect of each level of factors A and B.	01

$Experiments$	$FACTORS$			$Result$ ($quality\ characteristics$)
	A	B	C	
$Y1$	1	1	1	30
$Y2$	1	2	2	25
$Y3$	2	1	2	34
$Y4$	2	2	1	27

PART-B

2	a	Illustrate how you would apply the principles of DOE in conducting a full factorial experiment.	08
	b	How is an off-line quality control technique different from that of online? State some of the tools/techniques under each category. If you were asked to improve the product quality, which on would you prefer? Justify.	08

OR

3	a	Consider the step turning operation on a lathe. Construct the P-diagram for this operation.	04
	b	A pert dimension on a power tool is specified as 32.25 ± 0.25. Company records show ± 0.25 exceeded and 75% of the returned for replacement. Cost of replacement is $Rs.12{,}500$. Determine k and Quality Loss Function.	06
	c	With neat sketches, explain the variations in Taguchi's Quality Loss Function.	06
4	a	A chemist wishes to test the effect of four chemical agents on the strength of a particular type of cloth. Because there might be variability from one bolt to another, the chemist decides to use a randomized block design, with the bolts of cloth considered as blocks. She selects five bolts and applies all four chemicals in random order to each bolt. The resulting tensile strengths follow:	

$Chemical$	$Bolt$				
	1	2	3	4	5
1	73	68	74	71	67
2	73	67	75	72	70
3	75	68	78	73	68
4	75	71	75	75	69

Analyze the data from this experiment and draw appropriate conclusions. 10

b Illustrate the use of normal probability plots of residuals for checking the model adequacy. 06

OR

5 a The effect of five different ingredients (A, B, C, D, E) on reaction time of a chemical process is being studied. Each batch of new material is only large enough to permit five runs to be made. Furthermore, each runs requires approximately 1 ½ hours, so only five can be made in one day. The experimenter decides to run the experiment as a Latin square so that day and batch effects can be systematically controlled. She obtains the data that follows. Analyze the data from this experiment and draw conclusions.

$Batch$	Day				
	1	2	3	4	5
1	$A = 8$	$B = 7$	$D = 1$	$C = 7$	$E = 3$
2	$C = 11$	$E = 2$	$A = 7$	$D = 3$	$B = 8$
3	$B = 4$	$A = 9$	$C = 10$	$E = 1$	$D = 5$
4	$D = 6$	$C = 8$	$E = 6$	$B = 6$	$A = 10$
5	$E = 4$	$D = 2$	$B = 3$	$A = 8$	$C = 8$

10

b The final height of a biological specimen is recorded for the two factors time and medium in which it grows. Draw the interaction plot of time and medium.

		M	
		$Medium\ 1$	$Medium2$
T	$12\ hours$	21 23 20 22 28 26	25 24 29 26 25 27
	$18\ hours$	37 38 35 39 38 36	31 29 30 34 33 35

06

6 An engineer is interested in the effects of cutting speed (A), tool geometry (B) and cutting angle (C) on the life (in hours) of a machine tool. Two levels of each factor are chosen and three replicates of a 2^3 factorial design are run. The results are given below:

A	B	C	*Treatment Combination*	*Replicate I*	*Replicate II*	*Replicate III*
$-$	$-$	$-$	(1)	22	31	25
$+$	$-$	$-$	a	32	43	29
$-$	$+$	$-$	b	35	34	50
$+$	$+$	$-$	ab	55	47	46
$-$	$-$	$+$	c	44	45	38
$+$	$-$	$+$	ac	40	37	36
$-$	$+$	$+$	bc	60	50	54
$+$	$+$	$+$	abc	39	41	47

Analyze this experiment assuming that each replicate represents a block of a single production shift. 16

OR

7 a Consider the data given below. It was only possible to run a one-half fraction of the 2^4 design. Construct the design using data from replicate 1. Perform *ANOVA* for the same.

Treatment Combination	*Replicate I*	*Replicate II*	*Treatment Combination*	*Replicate I*	*Replicate II*
(1)	90	93	d	98	95
a	74	78	ad	72	76
b	81	85	bd	87	83
ab	83	80	abd	82	86
c	77	78	cd	99	90
ac	81	80	acd	79	75
bc	88	82	bcd	87	84
abc	73	70	$abcd$	80	80

10

b In a study, chemists are trying to determine the tellurium content of sea water. For this, four factors at two levels each are chosen and are presented in the table below.

i) If the chemist decides on a full factorial experiment, how many runs he has to conduct?

ii) If he wants to go for fractional experiment 2^{k-1}, what are the savings in the experiment runs?

iii) Build a fractional factorial experiment layout for 2^{k-1} experiments.

Factor/Levels	-1	$+1$
Na concentration	2.5	250
K concentration	3	280
Ca concentration	2.6	260
Mg concentration	2.3	240

06

8 a The design of a water pump is being studied to set the optimum design parameters. The evaluation criterion is water leakage during operation. The table below lists the factors and the levels at which they are tested.

Water Pump Design		
Factor	***Level 1***	***Level 2***
A: *Cover design*	*Cover* 1	*Cover* 2
B: *Gasket design*	*Gasket* 1	*Gasket* 2
C: *Front bolt torque*	*LSL*	*USL*
D: *Sealant*	*No*	*Yes*
E: *Surface finish*	*Rough*	*Smooth*
F: *Back bolt torque*	*LSL*	*USL*
G: *Torque sequence*	*Front, Back*	*Front, Back*

The outcome of the testing is measured in terms of the number of leaks in the pump assembly. The tests are conducted using an $L_8(2^7)$ orthogonal array and the raw results are tabulated below. Leakage was rated on a scale from $0 - -5$, where 0: no leaks, 5:extremely high leaking.

Trial No.	***Cover***	***Gasket***	***Front torque***	***Sealant***	***Surface torque***	***Back torque***	***Torque sequence***	***Test: leaking***
	A	***B***	***C***	***D***	***E***	***F***	***G***	
1	1	1	1	1	1	1	1	4
2	1	1	1	2	2	2	2	3
3	1	2	2	1	1	2	2	1
4	1	2	2	2	2	1	1	0
5	2	1	2	1	2	1	2	2
6	2	1	2	2	1	2	1	4
7	2	2	1	1	2	2	1	0
8	2	2	1	2	1	1	2	1

Analyze this problem using:
i) Ranking method
ii) Column effects method. 10

b Suppose you are using orthogonal array approach for an experiment and you have come across the following scenarios:
i) One control factor determines the appropriate choice of other control factors
ii) Turn a 2 level factors into 4 level factor
iii) Turn a 2 level factor into a 3 level factor

For each of the scenarios, state how you would proceed with the experimentation. Provide examples. 06

OR

9 a An arc welding experiment performed by railway cooperation in 1959. Nine variables were studied as shown in the table:

Factor	***Lower Level***	***Higher Level***
$A -$ *Kind of welding*	$J100$	$B17$
$B -$ *Drying method of rods*	*No drying*	*One − day*
$C -$ *welded material*	$SS41$	$SB35$

b			

Factor	***Lower Level***	***Higher Level***
$D-thickness\ of\ welded\ material$	$8mm$	$12mm$
$E-angle\ of\ welding\ device$	$70\ deg.$	$60deg.$
$F-opening\ of\ welding\ device$	$15mm$	$3mm$
$G-current$	$150A$	$130A$
$H-welding\ methods$	$Weaving$	$Single$
$I-preheating$	$No\ preheating$	$150deg.,preheating$

Along with these nine variables, following interactions are also of interest: AG, AH, GH and AC

Select the suitable orthogonal array. Construct the OA layout as well as physical layout for the experiment. 10

The main of the experimentation at Electronica Machine Tools is to optimize and stabilize the EDM performance characteristics namely

i) Material Removal Rate (MRR)

ii) Percent Electrode Wear (EW).

The control factors identified for the experimentation and their levels are provided below.

$Control\ Factors$	$Levels$		
	1	2	3
$Pulse-On-Time$	$150\mu sec$	$200\mu sec$	$500\mu sec$
$Gap\ current$	$30amp$	$34amp$	$50amp$
$Bi-Pulse\ Current$	$0amp$	$1amp$	$3amp$

$L9\ OA$ was selected for the experimentation. Do you think the selected array is the right one? If yes, draw the physical layout for the experimentation. 06

10

In the question 9b, suppose the experimenter continues with the $L9\ OA$, and gets the following data for MRR (with four repetitions)

Expt. No.	***4 Repetitions or Measurements for each expt.***			
1	168.3	169.2	161.2	161.1
2	221.4	220.5	214.2	215.3
3	318.3	317.7	312.4	310.9
4	192.4	191.5	188.7	187.1
5	238.2	239.7	233.9	231.6
6	312.6	311.2	307.3	308
7	198.4	197.1	192.8	191.9
8	181.1	182.3	178.9	177.4
9	325.8	324.4	317.8	316.3

The following data are the results (4 repetitions) of $L9\ OA$ experimentation for $EW\%$

Expt. No.	***4 Repetitions or Measurements for each expt.***			
1	15.2	16.5	12.2	11.6
2	7.2	7.4	6.4	6.6
3	11.2	11.3	10.6	10.9
4	2.6	2.6	2.3	2.5
5	4.2	4.3	3.8	3.7
6	15.3	15.4	14.9	15.2
7	0.65	0.7	0.5	0.6
8	7.3	7.2	6.8	6.8
9	2	1.9	1.5	1.4

		a) Identify the appropriate *SN* ration for the problem. b) Compute the *SN* ratio for each experimental run and tabulate them. c) Plot the *SN* ratio graphs and identify the optimal levels for the factor.	16
		OR	
11	a	Indicate under what conditions you would use each of these designs: i) Factorial Design ii) Taguchi Method iii) Random Design.	06
	b	Illustrate the application of steps in robust design for solving any problem related to gear cutting operation. (Assume the objective function and response suitably)	10

Project Planning and Control

V Semester SEE Question Papers

Course codes: 12IM5B4

- Nov/Dec 2017 (12IM5B4)
- Fast Track Jul 2017 (12IM5B4)
- Nov/Dec 2016 (12IM5B4)

12IM5B4

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-17

Industrial Engineering and Management

PROJECT PLANNING AND CONTROL (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

8. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
9. Answer FIVE full questions from Part B.
10. Use of statistical tables permitted.

PART-A

1	1.1	List the tools for identifying investment opportunities.	02
	1.2	State two entry barriers for threats of new entrants.	02
	1.3	The following information is available on quantity demanded and income level: $Q_1 = 60, Q_2 = 70, I_1 = 1000$ & $I_2 = 1200$. What is the income elasticity of demand?	02
	1.4	Given an actual demand of 60 for a period when forecast of 70 was anticipated and an alpha of 0.3, what would be the forecast for the next period?	02
	1.5	State any two sources of risk.	02
	1.6	What are the limitations of social cost benefit analysis?	02
	1.7	If project A has a net present value NPV of $Rs.30{,}000$ and project B has an NPV of $Rs.50{,}000$, what is the opportunity cost if project B is selected?	02
	1.8	Where the concept of $BOOT$ (Build, Own, Operate and Transfer) is used?	02
	1.9	How an effective monitoring can help in the progress of the project.	02
	1.10	Explain the concept of network technique.	02

PART-B

2	a	Explain clear articulation of objectives with respect to stimulating the flow of ideas.	10
	b	Explain any two aspects to be considered for corporate appraisal.	06
		OR	
3	a	Explain the Porter Model with a neat block diagram for identifying investment opportunities.	10
	b	Discuss any three aspects considered for preliminary screening of any project.	06
4	a	With an example, analyze the steps in conducting a sample survey for a project.	06

	b	During the past 8 quarters, the port of Baltimore has unloaded large quantities of grain ($\alpha = 10$). The first quarter forecast was 175. Find the forecast for the 9th quarter using exponential smoothing. *Quarter* | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 *Actual* | 180 | 168 | 159 | 175 | 190 | 205 | 180 | 182	10
		OR	
5	a	Discuss the sources of uncertainty of demand forecasting with respect to environmental changes.	10
	b	Differentiate between Qualitative Methods and Quantitative Methods.	06
6	a	Explain Hillier Model for uncorrected cash flow as well as perfectly correlated cash flows.	08
	b	A capital investment whose net present value has the following distribution. Calculate range, standard deviation, coefficient of variation and semi-variance. *Net Present value* | *Probability* 200 | 0.3 600 | 0.5 900 | 0.2	08
		OR	
7	a	State the five stages of appraisal in the *UNIDO* method as described in the Guide to Practical Project Appraisal.	10
	b	In a project, the income gained/lost by four groups viz., Group1 $= Rs.1000000$, Group 2 $= Rs.5000000$, Group 3 $= Rs.-2000000$ and Group 4 $= Rs.-4000000$. The marginal prosperity to save these four groups are $0.05, 0.10, 0.20$ & 0.40. calculate the impact on savings of the project.	06
8	a	List the important key project parties and state their fundamental responsibilities of any two.	10
	b	Discuss with help of an example the following risks in a private infrastructure projects: i) Construction Risk ii) Market Risk iii) Regulatory Risk.	06
		OR	
9	a	Define public private partnership and explain the relevance of public private partnership.	08
	b	State the recommendations of the committee on infrastructure financing.	08
10	a	Enumerate the characteristics of sound project organization for successful project implementations.	06

	b	A project was began on 1st Jan 2014 and was expected to be completed by 30th Sep 2014. The project is being reviewed on 30th June 2014 when the following information has been developed $Budgeted\ cost\ for\ work\ schedule$ — $Rs.1{,}500{,}000$ $Budgeted\ cost\ for\ work\ performed$ — $Rs.1{,}400{,}000$ $Actual\ cost\ of\ work\ performed$ — $Rs.1{,}600{,}000$ $Budgeted\ cost\ for\ total\ work$ — $Rs.2{,}500{,}000$ $Estimated\ cost\ for\ completion$ — $Rs.1{,}200{,}000$ Determine the following: i) Cost variance ii) Schedule variance in cost terms iii) Cost performance index iv) Schedule performance index v) Estimated cost performance index.	10
		OR	
11	a	Explain the steps of Fulkerson's Rule for naming the node in a network diagram.	04
	b	A project consists of the following activities with the duration in days $Activity$ — A — B — C — D — E — F — G — H — I $Precedence$ — - — A — A — B,C — A — D,E — C — F,G — H $Duration$ — 10 — 4 — 12 — 10 — 9 — 8 — 7 — 9 — 6 i) Draw the network of the above project. ii) Identify the critical path and determine the project duration iii) Calculate early and late start, early and late finish, independent float, free float for all the activities.	12

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Fast Track Examinations July-17

Industrial Engineering and Management

PROJECT PLANNING AND CONTROL (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	A series of activities to create a unique product or service by a specific data is _____.	01
	1.2	State any two basic competitive forces used in Porter model.	02
	1.3	List different performance indicators _____.	02
	1.4	RAT & CAT in project management stands for _____ and _____.	02
	1.5	List the market survey techniques.	02
	1.6	State the three time values of the $PERT$ time estimation.	02
	1.7	Define free float and independent float.	02
	1.8	Represent the network for the precedence relationship activities such that C & D cannot stand until the activities A and B are completed.	02
	1.9	What is sensitivity analysis?	02
	1.10	What is mante carlo simulation?	02
	1.11	State any two sources of work in project context.	01

PART-B

2	a	Discuss the various factors to be considered for project preliminary screening.	08
	b	Analyze the various types of project rating indices with examples.	08
		OR	
3	a	List and explain the various aspects for evaluating the scouting of project ideas.	08
	b	Identify the various sectors considered for monitoring the project environment. Describe each one of them.	08
4	a	Enumerate the factors to be considered in locating a project site with an example.	08
	b	Explain the characteristics of a good demand forecasting method.	08
		OR	

5	a	What are the various methods to determine the demand forecasts for a consumable product?	08
	b	With an example discuss the product mix flexibility for automobile components.	08
6	a	Explain the various types of risks associated in a project.	08
	b	Discuss the $UNIDO$ approach to analyze the social cost benefit analysis.	08
		OR	
7	a	Discuss the various projects selected under the risk.	08
	b	List and explain the activities involved in risk analysis.	08
8	a	If cash flow of different years are perfectly uncorrelated how is the standard duration of NPV defined by the Hiller model?	08
	b	Explain the objectives of shareholders agreement and EPC contract.	08
		OR	
9	a	Evaluate the Financial challenges of a solar power project.	08
	b	How would you evaluate the appropriateness of a technology?	08
10	a	List the information required for preparing the project implementation schedule.	08
	b	Differentiate between $PERT$ and CPM.	08
		OR	
11	a	What are the prerequisites for successful project implementation?	08
	b	Discuss the various forms of project organization structures.	08

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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

V Semester B. E. Examinations Nov/Dec-16

Industrial Engineering and Management

PROJECT PLANNING AND CONTROL (ELECTIVE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

8. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
9. Answer FIVE full questions from Part B.

PART-A

1	1.1	Periodic $SWOT$ analysis facilitates the _____ of ideas.	01
	1.2	Porter model, life cycle approach, experience curve are the tools for identifying _____ opportunities.	01
	1.3	List the components of time series method.	02
	1.4	List different performance indicators of project.	02
	1.5	Find the co-efficient of variation when standard deviation is 320 and expected value is 560.	02
	1.6	A common way to modify the risk of an investment is to changes the proportion of _____ & cost.	02
	1.7	State any two important stages in life cycle of project.	02
	1.8	The following information is available on quantity demanded and income level: $Q_1 = 40, Q_2 = 50, I_1 = 1000$ and $I_2 = 1500$. What is the income elasticity of demand?	02
	1.9	In $PERT$ the span of time between the _____ and _____ time estimates of an activity is 6σ.	02
	1.10	Network is the graphical display of a project that contains both _____ and _____.	02
	1.11	List any four risks in managing infrastructure projects.	02

PART-B

2	a	List the important aspects to be considered under corporate appraisal. Explain.	08
	b	Explain why preliminary screening is required to eliminate ideas which prima-facie are not promising.	08
		OR	
3	a	Industry economists believe all the manufactured product evolves through different product life cycle. Discuss the stages in the product life cycle.	08
	b	Analyze various types of project rating index with illustrations.	08

4	a	With a neat block diagram explain key steps in Market and Demand analysis and their inter-relationship.	10
	b	Present the considerations that have a strong influence on the choice of location and site.	06
		OR	
5	a	List the information required for preparing the project implementation schedule.	06
	b	Material inputs are classified into four broad categories. Explain each of them.	10
6	a	Explain the sources of risk. Give an example to trace the overall risk of the project to their source.	08
	b	As per the *UNIDO* method, how are the following valued? i) Tradable inputs and outputs ii) Non-tradable inputs and outputs iii) Labour iv) Foreign exchange.	08
		OR	
7	a	Illustrate with an example Net economic benefit of a bridge construction project.	06
	b	What are the risks in project management a firm deals with? Explain the types of risks involved in it with examples.	10
8	a	Explain the recommendations of the committee on infrastructure financing in project management.	06
	b	Explain financing a power project with respect to electricity generation projects.	10
		OR	
9	a	What is Public private partnership (*PPP*)? Explain the relevance of *PPP* in India.	06
	b	Apart from the license itself, how project financiers in telecoms sector evaluate the projects?	10
10	a	Widgetco is about to introduce a new product. One unit of this product is produced by assembling subassembly 1 and subassembly 2. Before production begins on either subassembly, raw materials must be purchased and workers must be trained. Before the sub-assemblies can be assembled into final product, the finished subassembly 2 must be inspected. A list of activities, their predecessors, and their durations is given in the following table.	

Activity	*Predecessors*	*Duration(days)*
$A-$ *Train workers*	---	6
$B-$ *purchase raw materials*	---	9
$C-$ *Make subassembly* 1	A, B	8
$D-$ *make subassembly* 2	A, B	7
$E-$ *inspect subassembly* 2	D	10
$F-$ *Assemble Subassemblies*	C, E	12

		i) Draw a project diagram (network) for this project. ii) What is the total project time? iii) What is the critical path?	10
	b	Write short notes on the following with respect to human relations in project setting: i) Authority ii) Orientation iii) Motivation.	06
		OR	
11	a	Consider the following data of a project.	
	b	What is bar chart? What are the advantages of bar chart?	06

$Activity$	$Predecessor$	$Duration(weeks)$		
		a	m	b
A	-	3	5	8
B	-	6	7	9
C	A	4	5	9
D	B	3	5	8
E	A	4	6	9
F	C, D	5	8	11
G	C, D, E	3	6	9
H	F	1	2	9

i) Construct the project network.
ii) Find the expected duration and variance of each activity.
iii) Find the critical path and the expected project completion time. 10